

UNIT – I

FUNDAMENTALS OF IoT: Evolution of Internet of Things, Enabling Technologies, IoT Architectures, oneM2M, IoT World Forum (IoTWF) and Alternative IoT models, Simplified IoT Architecture and Core IoT Functional Stack, Fog, Edge and Cloud in IoT, Functional blocks of an IoT ecosystem, Sensors, Actuators, Smart Objects and Connecting Smart Objects.

What is IoT ?

➤ Goal of IoT:

➤ Connect the unconnected

Objects that are not currently joined to a computer network-Internet, will be connected so that they can communicate and interact with people and other objects.

- IoT is a technology transition in which the devices will allow us to sense and control the physical world by making objects smarter and connecting them through an intelligent network.
- When objects and machines can be sensed and controlled remotely by across a network, a tighter integration between physical world and computers are enabled. This allows enablement of advanced applications.

In the Internet of Things, all the things that are being connected to the internet can be put into three categories:

- Things that collect information and then send it.
- Things that receive information and then act on it.
- Things that do both.

1. Collecting and Sending Information

Sensors could be temperature sensors, motion sensors, moisture sensors, air quality sensors, light sensors.

These sensors, along with a connection, allow us to automatically collect information from the environment which, in turn, allows us to make more intelligent decisions.

On a farm, automatically getting information about the soil moisture can tell farmers exactly when their crops need to be watered. Instead of watering too much (which can be an expensive over-use of irrigation systems) or watering too little (which can be an expensive loss of crops), the farmer can ensure that crops get exactly the right amount of water. This enables farmers to increase their crop yield while decreasing their associated expenses.

2.Receiving and Acting on Information

The machines get information and then acting.

Ex: printer receives a document and it prints it.

A car receives a signal from your car keys and the doors open. The examples are endless.

Whether it's a simple as sending the command “turn on” or as complex as sending a 3D model to a 3D printer, we know that we can tell machines what to do from far away. Things that collect information and send it, but also receive information and act on it.

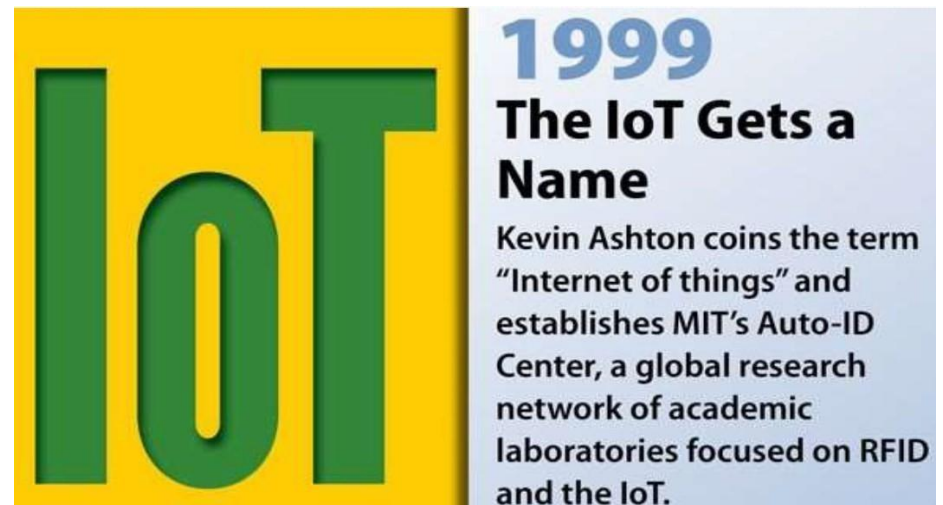
3.Doing Both: The Goal of an IoT System

Farming example: The sensors can collect information about the soil moisture to tell the farmer how much to water the crops, but you don't actually need the farmer. Instead, the irrigation system can automatically turn on as needed, based on how much moisture is in the soil.

What is IoT ?

➤History:

- Kevin's Explanation:
- IoT involves the addition of senses to computers.
- In the 20th century, computers were brains without senses.
- In the 21st century, computers are sensing things for themselves.



The **Internet of Things (IoT)** refers to a network of physical objects, or "things," that are embedded with sensors, software, and other technologies that enable them to connect and exchange data with other devices and systems over the internet

"Things":

These can be anything from simple sensors to complex machines, vehicles, appliances, and even living organisms (like people with wearable technology).

The **basic goal** of IoT is to **connect the unconnected things** so that they can communicate and interact with people and other objects

IoT is not single technology, but it is good to view it as an **umbrella of various concepts, protocols, and technologies**

Sensors:

These devices collect data about the environment or the "thing" itself, such as temperature, location, or movement.

Connectivity:

This allows the "things" to communicate with each other and with other systems, typically over the internet.

Data Exchange:

The core function of IoT is the ability to gather data and share it, often in real-time, to facilitate various applications.

Automation:

By connecting devices and enabling data exchange, IoT allows for automated processes and tasks, reducing the need for human input.

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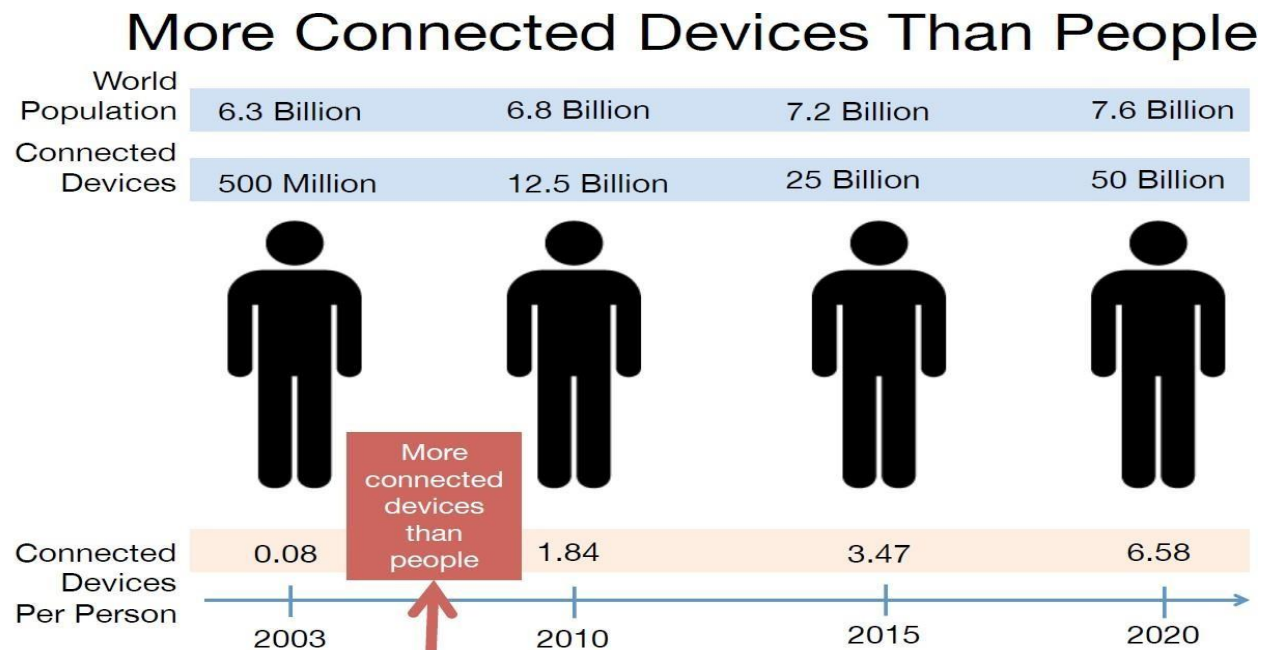
Fundamentals of IoT

Genesis of IoT (Evolution of Internet of Things)

- The Internet of Things (IoT) era began around 2008 - 2009 when more devices were connected to the internet than people.
- Kevin Ashton coined the term "Internet of Things" in 1999.
- IoT enables computers to sense things on their own, marking a major technology shift.

Genesis of IoT:

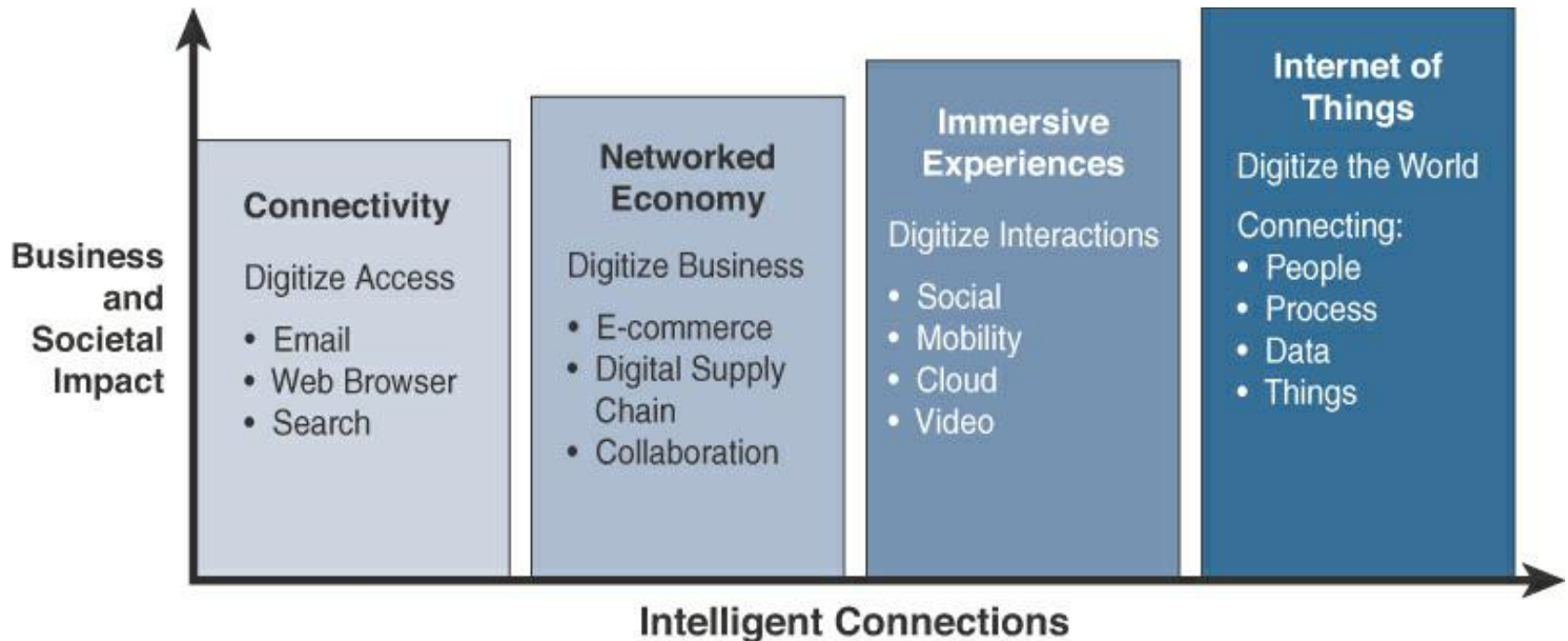
- The age of IoT is started in 2008 and 2009. In these years, more things connected to the Internet than people in the world.



[Source: Cisco IBSG, April 2011]

Evaluation of Internet

The evolution of the Internet can be categorized into four phases.



- IOT explains a way of connecting of different set of things and objects to the internet. while digitization focuses on the way of connecting objects, things, people with the data which emerges as a business process.
- Every field is marching towards digitization. For example photography, video retail industry, transportation (eg.ola and Uber). Home automation is an good example of digitization where sensors play a vital role (eg.Nest).

These four phases of internet further defined as

Internet Phase	Definition
Connectivity (Digitize Access)	This phase connected people to email, web services and search, so that information is easily accessed.
Networked Economy (Digitize Business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business.
Immersive Experiences (Digitize Interactions)	This phase extended the Internet Experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved to Cloud.
Internet of Things (Digitize the World)	This phase is adding connectivity to Objects and machines to the world around us to enable new services and experiences. It is connecting the unconnected.

Connectivity Digital access (Email)	Networked Economy Digital business (E-commerce)	Immersive Experience Digital Interactions (social, cloud)	Internet of Things Digitize the world Connect People Data Things process
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Evolutionary Phases of the Internet

- Each phase of evolutionary phases builds on the previous one.
- With each subsequent phase, more value becomes available for businesses, governments and society in general.

Internet Phase: first Phase	Connectivity(Digitize Access)
➤ Began in the mid 1990s. ➤ Email and getting Internet were luxuries for universities and corporations. ➤ Dial-up modems and basic connectivity were involved. ➤ Saturation occurred when connectivity and speed was not a challenge. ➤ The focus now was on leveraging connectivity for efficiency and profit.	

➤ Evolutionary Phases of the Internet

Internet Phase: Second Phase	Networked Economy (Digitize Business)
➤ E-Commerce and digitally connected supply chains become the rage. ➤ Caused one of the major disruptions of the past 100 years.. ➤ Vendors and suppliers became closely interlinked with producers. ➤ Online Shopping experienced incredible growth . ➤ The economy become more digitally intertwined as suppliers, vendors and consumers all became more directly connected.	

➤ Evolutionary Phases of the Internet

Internet Phase: Third Phase	Immersive Experiences (Digitize Interactions)
➤ Immersive Experiences, is characterized by the emergence of social media, collaborations and widespread mobility on a variety of devices. ➤ Connectivity is now pervasive, using multiple platforms from mobile phones to tablets to laptops and desktop computers. ➤ Pervasive connectivity enables communications and collaboration as well as social media across multiple channels via email, texting, voice and video. ➤ Person to person interactions have become digitized.	

➤ Evolutionary Phases of the Internet

Internet Phase: Forth(last) Phase	Internet of Things (Digitize the World)
➤ We are in beginning of the IoT phase. ➤ 99% of things are still unconnected. ➤ Machines and objects in this phase connect with other machines and objects along with humans. ➤ Business and society are using and experiencing huge increase in data and knowledge. ➤ Increased automation and new process efficiencies, IoT is changing our world to new way.	

IoT and Digitization

- At a high level, IoT focuses on connecting things such as objects and machines, to a computer network, such as the Internet.
- Digitization encompasses the connection of things with the data they generate and the business insights that result.

Example: Wi-Fi devices in Malls detecting customers, displaying offers, based on the spends, mall is segregated, changes the location of product displays and advertising.

- Digitization: It is the conversion of information into a digital format.

IoT and Digitization

Example:

1.Digital camera- No films used, mobile phones with camera.

Digitization of photography changed experience of capturing images.

2.Video rental industry and transportation , no one purchases video tapes or DVDs. With digitization, everyone is streaming video content or purchasing the movies as downloadable files.

3.Transportation- Taxi Uber, Ola use digital technologies.

4.Home Automation – Popular product: Nest – sensors determine the climate and connects to other smart objects like smoke alarm, video camera and various third party devices.

➤ Ex: Nest Learning Thermostat, Nest Protect and Nest Cam enable users to set heating, check in-house cameras and smoke levels from their smartphone or another Wi-Fi device.

Internet of Things – Enabling Technologies

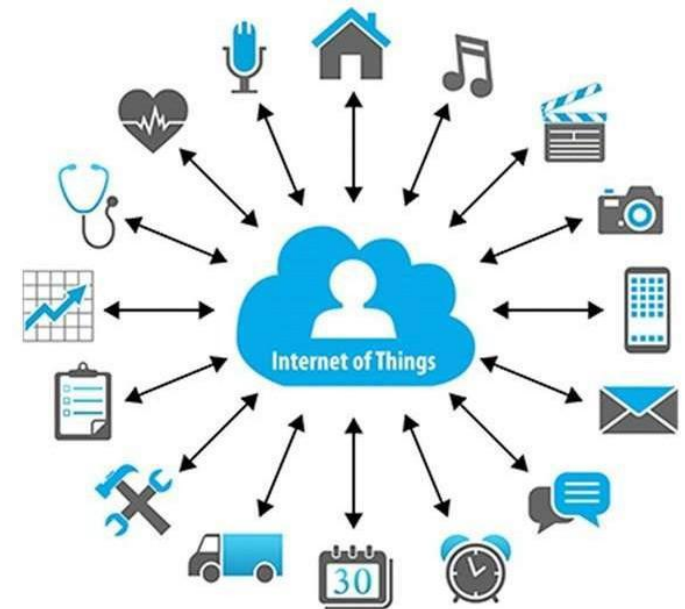
Internet of Things – Enabling Technologies

IoT (Internet of Things) relies on several key enabling technologies

The Internet of Things (IoT) is a growing network of connected objects with IP addresses, enabling internet connectivity and communication between devices and systems.

Key Aspects

- **Extended Connectivity:** Beyond traditional devices (PCs, mobile phones, tablets) to various everyday objects.
- **Interaction with Environment:** Devices utilize inbuilt technology to sense and interact with the external environment via the Internet.
- **Massive Scale:** 50 billion devices connected worldwide via IoT technology.



IoT enables communication between objects and systems, transforming the way we live and work.

IoT (Internet of Things) relies on several key enabling technologies

IoT Enabling Technologies

- **Wireless Sensor Network**



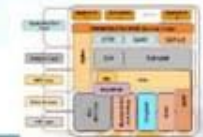
- **Cloud Computing**



- **Big Data Analytics**



- **Communication Protocols**



- **Embedded Systems**



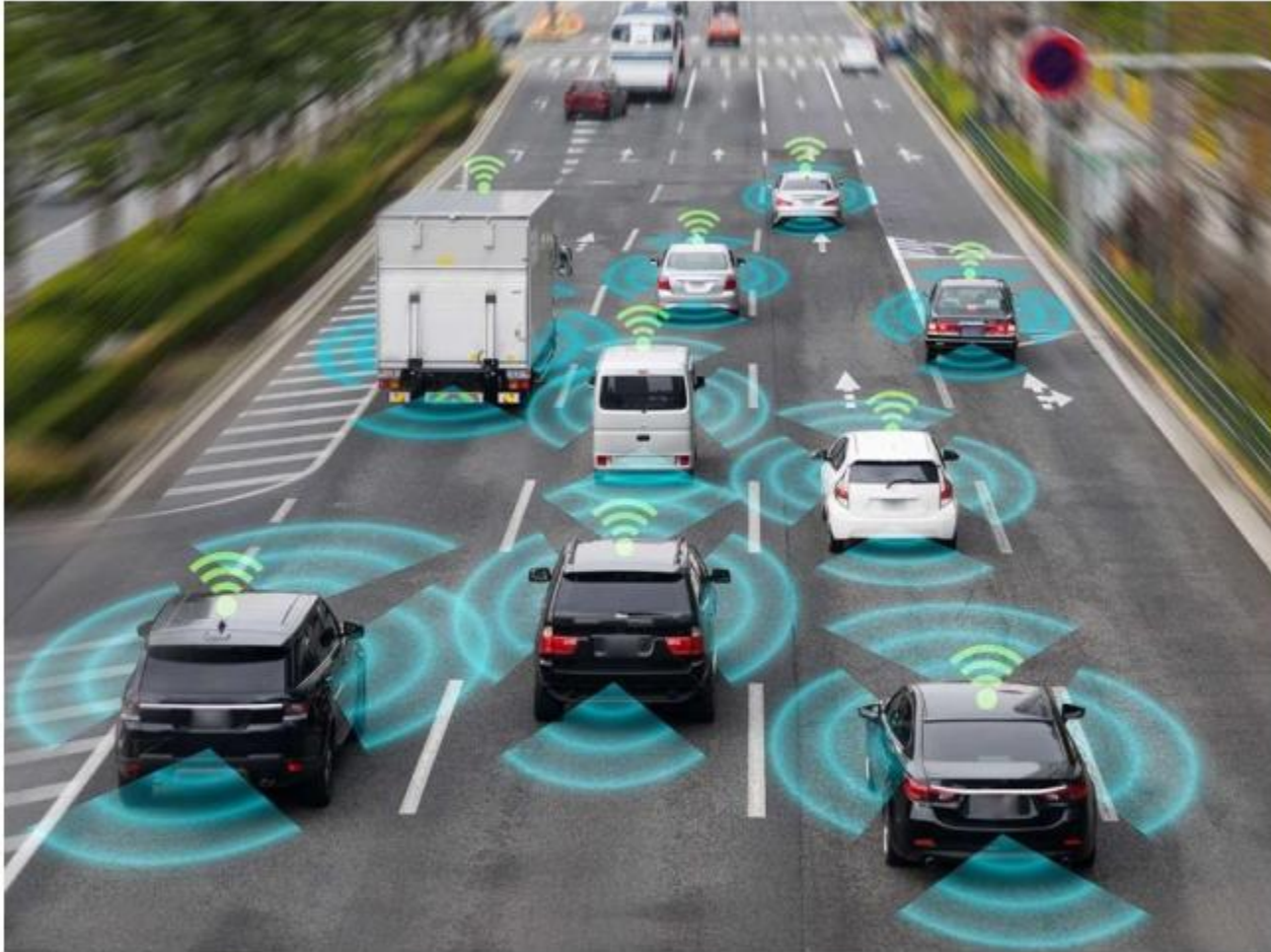
IOT has grown in many fields which have created an impact on **connected roadways, connected factory, smart connected building and smart creatures.**

Connected Roadways:

Self driving cars are introduced by Google which uses the **sensors** to **monitor the operating conditions of the vehicles**. The challenges encountered by the smart roadways are safety, mobility and environment of the vehicle. This smart roadway has **reduced the possibility of congestion due to traffic**. As vehicles are connected with internet it gets information regarding the weather conditions on the road and it makes necessary decisions.



Connected Roadways:



connected roadways

Connected factory

The **main challenges** facing manufacturing in a factory environment today include the following:

1. Accelerating new product and service introductions to meet customer and market opportunities
2. Increasing plant production, quality, and uptime while decreasing cost
3. Mitigating unplanned downtime (which wastes, on average, at least 5% of production)
4. Securing factories from cyber threats
5. Decreasing high cabling and re-cabling costs (up to 60% of deployment costs)
6. Improving worker productivity and safety

Industries are retooling their factories with advanced technologies and architectures to resolve these problems and boost manufacturing flexibility and speed.

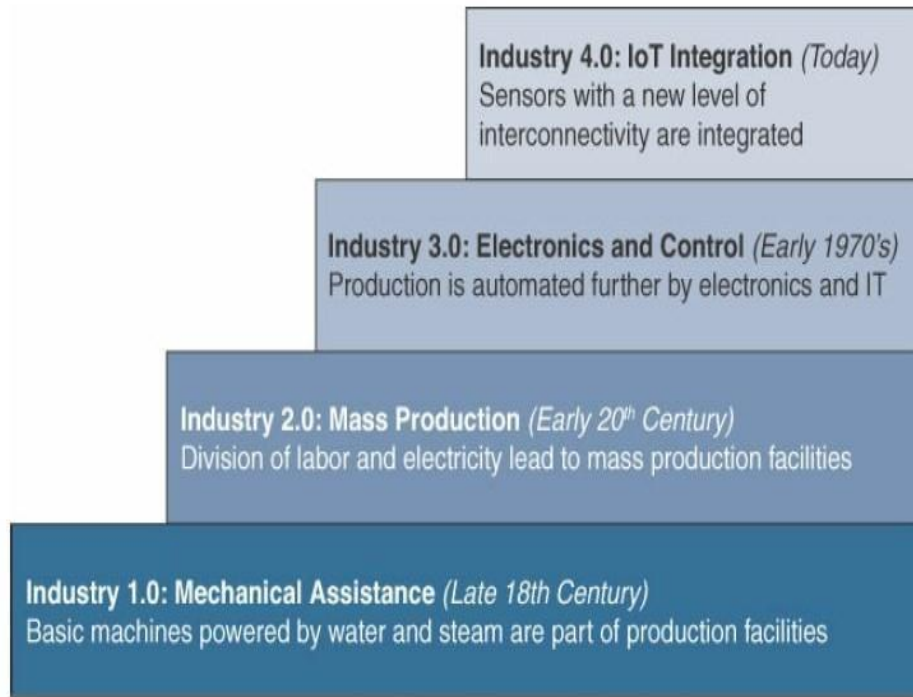
There are four industrial revolution namely

Industry 1.0, (It deals with the mechanical innovations),

Industry 2.0 (Production level was increased by Companies in this revolution),

Industry 3.0 (automation by electronics and information technology),

Industry 4.0 (IOT sensors integrated with the **environment**)



The Four Industrial Revolutions



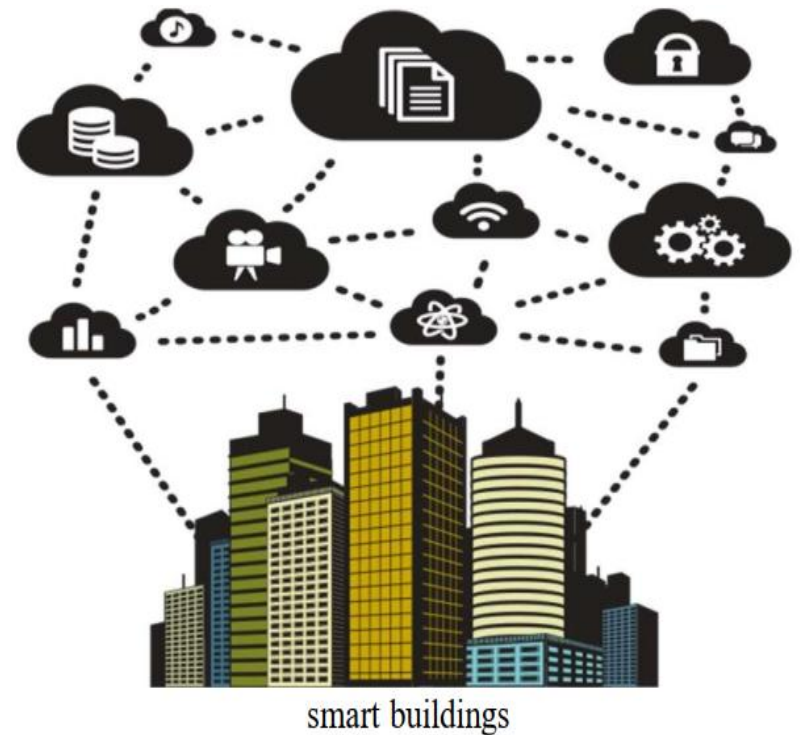
connected factories

Smart connected building

Building management systems are used in building smart building systems which controls the ventilation, temperature inside the room. Digital ceiling will have an Wi-Fi access point, sensors, surveillance cameras, HVAC, Lighting control.

HVAC (Heating, ventilation and air conditioning system)

BMS (building management system)
BAS (Building automation system),
BACnet (Building automation and control network) are few smart building systems are incorporated with the help of IOT.

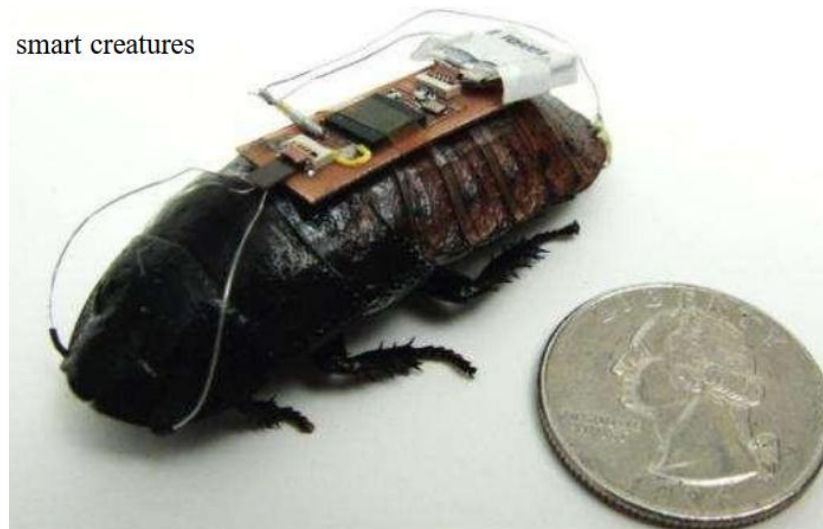


Smart creatures:

Smart devices are placed on the animals, through which we gain access to the information and behavior of the animal in an controlled environment. This helps in finding out the diseases and its cause.

Sensors are placed on the roaches to predict the disaster events which are going to occur.

The dutch company laid device on cows war, North Carolina state university experimented with cockroaches



IoT challenges

IoT challenges:

There are significant challenges faced by IOT Industry. They are scalability of the system, security and privacy of the system, interoperability and in handling of big data and data analytics.

Scale: It refers to the number of people connected to a IOT system. For example: Through IPV6 millions of devices are connected to the network.

Security: Many devices connected together so any device can attack the other devices connected to each other. So security is considered to be major challenge in IOT industry.

Privacy: The sensors gather data and provide health information, shopping patterns to business people. They may even misuse that data.

IoT challenges:

Big data and data analytics. Huge amount of data is collected from various devices connected to internet. Handling large set of data becomes a major challenge.

Interoperability: Different protocols are used by different devices. So interoperability between devices occurs. Many Iot standards are developed to handle these issues and challenges.

IoT Architectural Drivers.

Challenges	Description	IoT Architectural Changes required
Scale	The massive scale of IoT endpoints (sensors) is far beyond that of typical IT networks.	• The IPv4 address space has reached exhaustion and is unable to meet IoT's scalability requirements. • Scale can be met only by IPv6. • IT networks continue to use IPv4 through features like Network Address Translation.
Security	IoT devices, especially those on wireless sensor networks(WSNs) are often physically exposed to the world.	• Security is required at every level of the IoT network. • Every IoT endpoint node on the network must be part of the overall security strategy and must support device level authentication and link encryption. • It must also be easy to deploy with some type of a zero – touch deployment model.

Challenges	Description	IoT Architectural Changes required
Devices and networks constrained by power, CPU memory and link speed	Due to the massive scale and longer distances, the networks are often constrained, lossy and capable of supporting only minimal data rates (10s of bps to 100s of kbps)	• New-last mile wireless technologies are needed to support constrained IoT devices over long distances. • The network is also constrained, i.e modifications need to be made to the traditional network-layer transport mechanisms.
The massive volume of data generated	The sensors generate the massive amount of data on daily basis, causing network bottlenecks and slow analytics in the cloud.	• Data analytics capabilities need to be distributed throughout the IoT network, from the edge to the cloud. • In traditional IT networks, analytics and applications typically run only in the cloud.

Challenges	Description	IoT Architectural Changes required
Support for legacy systems	An IoT network often comprises a collection of modern, IP capable end points as well as legacy , non-IP devices that rely on serial or proprietary protocols.	• Digital transformation is a long process that may take many years , and IoT networks need to support translation and / or tunneling mechanisms to support legacy protocols over standards-based protocols, such as Ethernet and IP.
The need for data to be analyzed in real time	Where as Traditional IT networks perform scheduled batch processing of data, IoT data needs to be analyzed and responded to in real – time.	• Analytics software need to be positioned closer to the edge and should support real-time streaming analytics. • Traditional IT analytics software (such as relational database or even Hadoop), are better suited to batch-level analytics hat occur after the fact.

IoT Network Architecture

IoT Network Architecture

- 1.Importance of Architecture: Careful planning and design are crucial for IoT networks, just like building a house.
2. Differences from IT Networks: IoT networks have unique challenges and requirements, such as scale, security, and constrained devices.

Key Challenges

1. Scale: IoT networks need to support millions of devices.
2. Security: IoT devices are vulnerable to cyber attacks and require robust security measures.
3. Constrained Devices: IoT devices have limited power, CPU, and memory.
4. Legacy Device Support: IoT networks need to support older devices with different protocols

IoT Architectures

1. oneM2M: A standardized architecture for IoT.
2. IoT World Forum: A framework for IoT architecture.

The one M2M IoT Standardized Architecture

Overview

- Global Initiative: oneM2M is a global initiative to promote efficient M2M communication systems and IoT.
- Common Services Layer: oneM2M aims to create a common services layer for IoT devices and applications.

The one M2M IoT Standardized Architecture

- M2M: Machine-to-Machine communication, standardized by ETSI
- Architecture: 3 layers
- Application Layer: Connectivity between devices and applications, standardizes business intelligent systems
- Service Layer: Represents physical network, includes management protocol, hardware, and middleware, supports Restful APIs
- Network Layer: Incorporates relationships between communication domains and end systems, various wireless and wired networking standards
- Key Features: Enables communication between devices and application servers, focuses on IoT services and applications (smart grid, smart city, smart vehicles)
- Benefits: Supports various industries (business domains, industrial automation, healthcare), enables smart applications
- Technology: Supports LoRaWAN, IEEE 802.15.4, IEEE 802.11ah, IEEE 1901, etc.

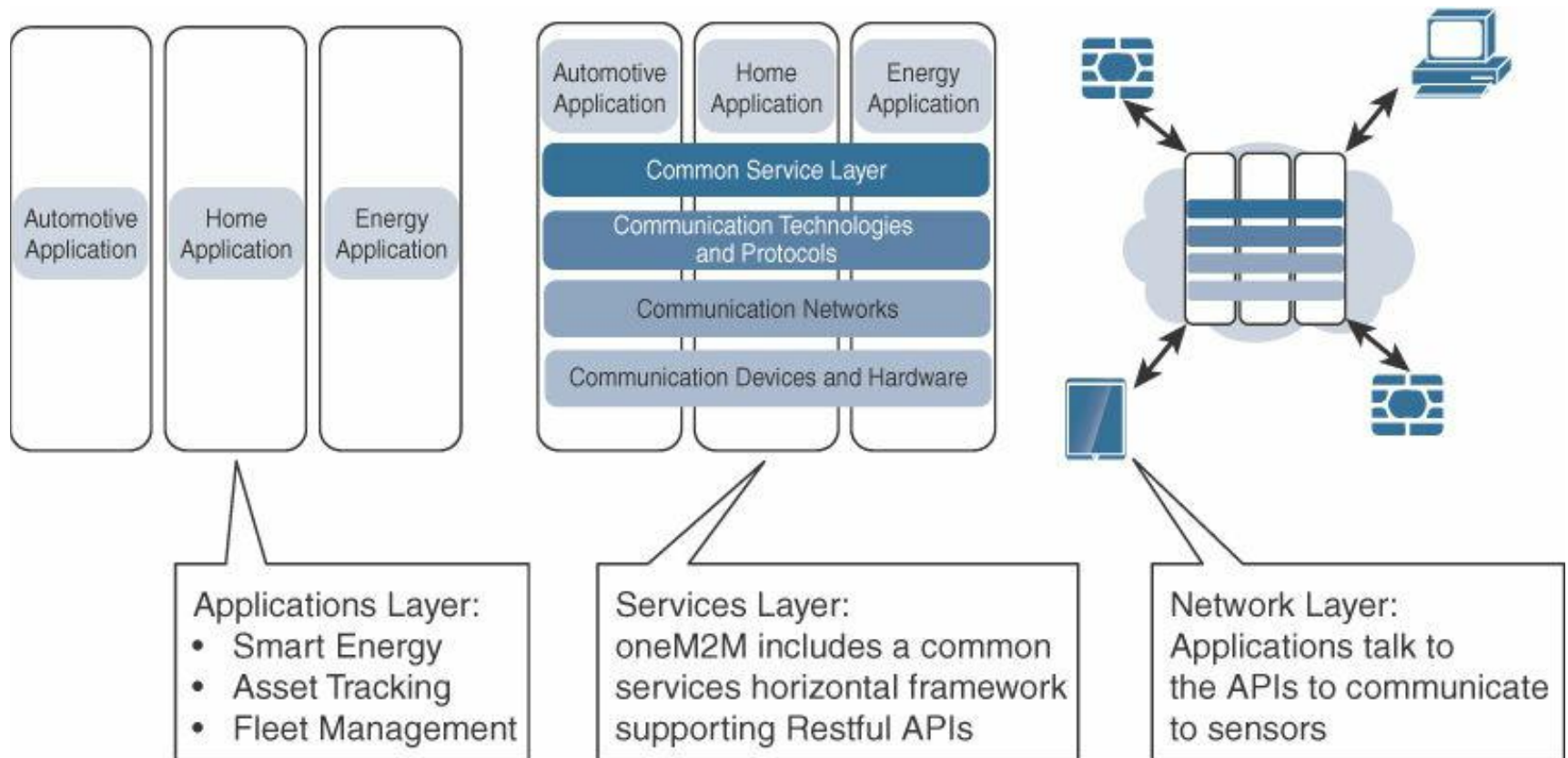


Fig: The Main Elements of the oneM2M IoT Architecture

Architecture of oneM2M

- Three Domains: The oneM2M architecture consists of three domains: Application Layer, Services Layer, and Network Layer.
- Application Layer: Focuses on connectivity between devices and applications, with standardized northbound API definitions.
- Services Layer: A horizontal framework that provides APIs and middleware for third-party services and applications.
- Network Layer: The communication domain for IoT devices and endpoints, including wireless and wired connections.

Benefits

- Interoperability: oneM2M promotes interoperability at all levels of the IoT stack.
- Standardization: oneM2M develops technical specifications for a common M2M Service Layer.

IoT World Forum (IoTWF) Architecture

- 7-Layer Reference Model: Published in 2014, it provides a technical perspective on IoT.
- IoTWF 7-Layer Architecture- Developed by Cisco, IBM, and Rockwell Automation in 2014.
- Key Features:
 - Edge computing, data storage, and access
 - Security across all 7 layers
 - Service provision from data center or cloud
- Benefits:
 - Simplifies IoT problems
 - Identifies relationships between technologies
 - Differentiates components from various vendors
 - Addresses interoperability issues
 - Deploys security at each layer
- This architecture provides a comprehensive framework for IoT solutions, ensuring security, interoperability, and efficient data management.

The IoT World Forum (IoTWF) Standardized Architecture:

- The IoT Reference Model defines a set of levels with control flowing from the center (this could be either a cloud service or a dedicated data center), to the edge, which includes sensors, devices, machines and other types of intelligent end nodes.
- In general, data travels up the stack, originating from the edge, and goes northbound to the center.
- Using this reference model, we are able to achieve the following:
 - Decompose the IoT problem into smaller parts
 - Identify different technologies at each layer and how they relate to one another
 - Define a system in which different parts can be provided by different vendors
 - Have a process of defining interfaces that leads to interoperability
 - Define a tiered security model that is enforced at the transition points between levels

IoT World Forum (IoTWF) Architecture

7-Layer Reference Model: Published in 2014, it provides a technical perspective on IoT.

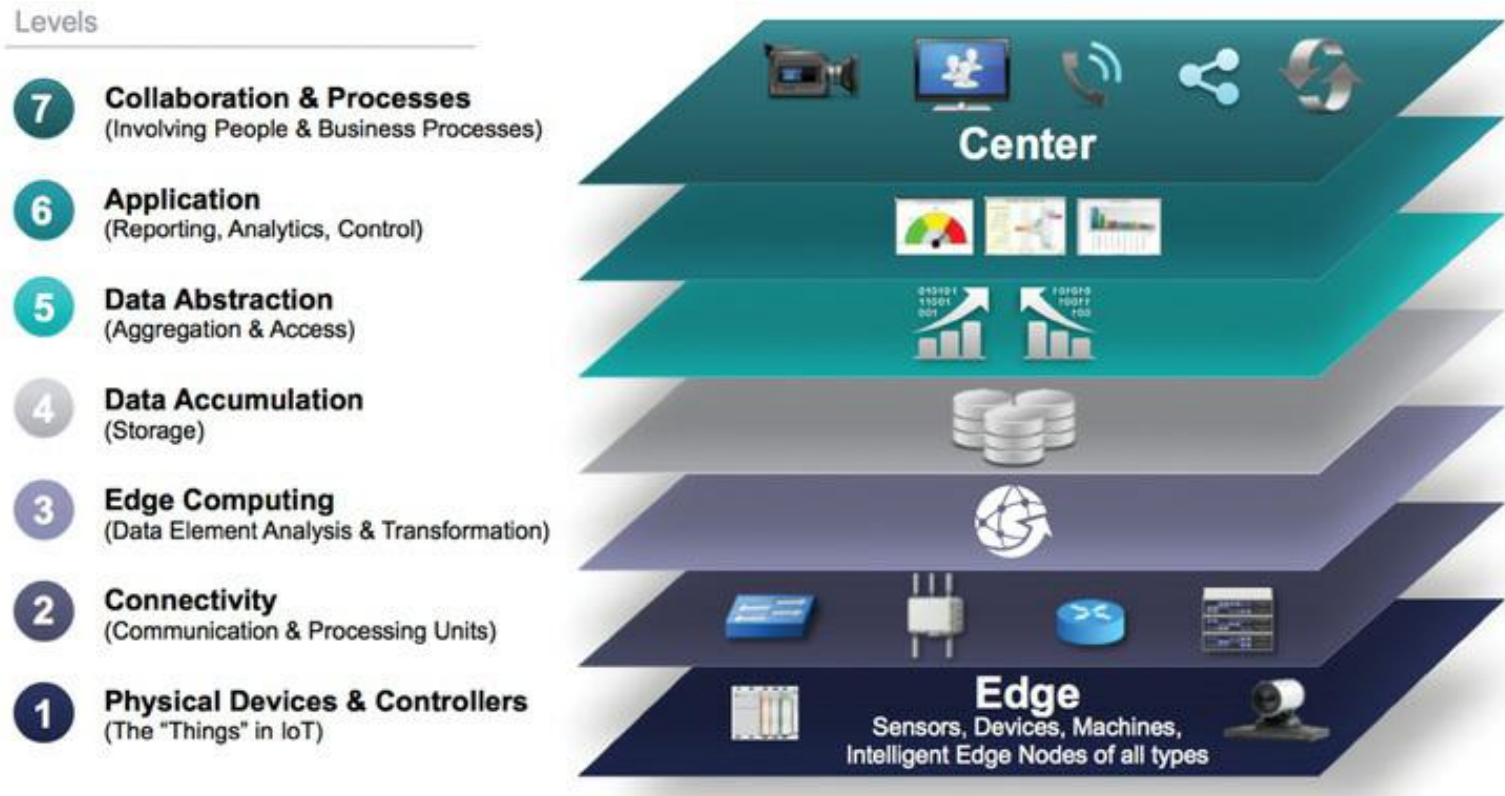


Fig: IoT Reference Model Published by the IoT World Forum

7 layers in the IoT Reference Model:

- 1. Physical Devices: Includes sensors, actuators, and other devices that interact with the physical world.
- 2. Connectivity: Encompasses communication protocols and networks (wired/wireless) for device-to-device and device-to-cloud interactions.
- 3. Edge Computing: Involves data processing and analysis at the edge of the network, closer to the devices.
- 4. Data Accumulation: Focuses on collecting, storing, and managing IoT data.
- 5. Data Abstraction: Provides APIs and interfaces for data access and manipulation.
- 6. Application: Includes software applications that utilize IoT data and insights.
- 7. Collaboration and Processes: Integrates IoT with business processes, workflows, and other systems.
- These layers work together to enable efficient, scalable, and secure IoT solutions.

IoT World Forum (IoTWF) Architecture

Layer 1: Physical Devices and Controllers Layer

- The first layer of the IoT Reference Model is the physical devices and controllers layer.
- This layer is home to the things in the Internet of Things, including the various endpoint devices and sensors that send and receive information.
- The size of these things can range from almost microscopic sensors to giant machines in a factory.
- Their primary function is generating data and being capable of being queried and/or controlled over a network.

Internet of Things Reference Model

1

**Physical Devices &
Device Controllers**
(The "Things" in IoT)

IoT "devices" are capable of:

- Analog to digital conversion, as required
- Generating data
- Being queried / controlled over-the-net

- Sensors, devices and controllers.



Layer 2: Connectivity Layer

- In the second layer of the IoT Reference Model, the focus is on connectivity.
- The most important function of this IoT layer is the reliable and timely transmission of data.
- More specifically, this includes transmissions between Layer 1 devices and the network and between the network and information processing that occurs at Layer 3 (the edge computing layer).
- The connectivity layer encompasses all networking elements of IoT and doesn't really distinguish between the last-mile network, gateway, and backhaul networks.

② **Connectivity**
(Communication and Processing Units)

Layer 2 Functions:

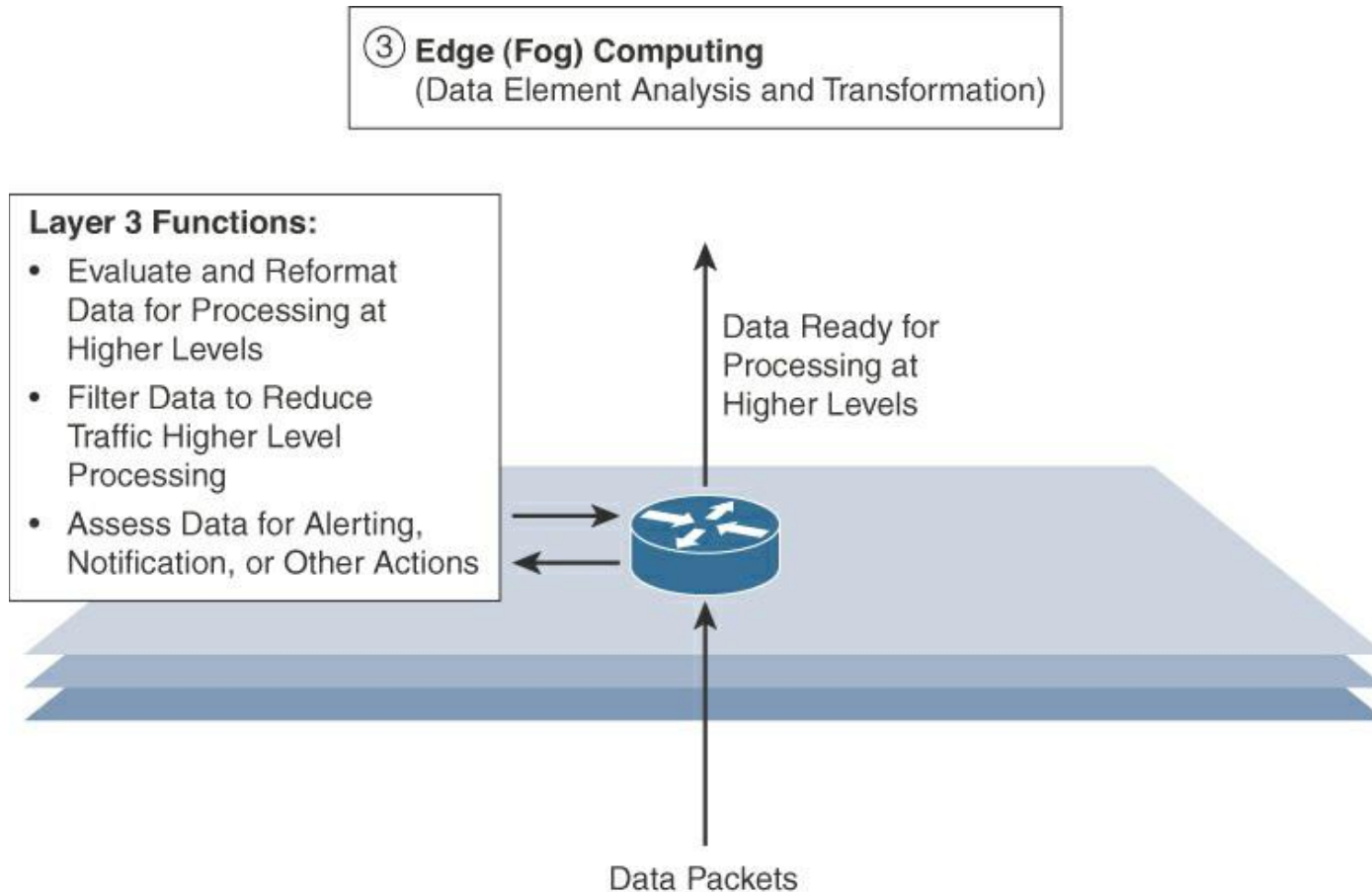
- Communications Between Layer 1 Devices
- Reliable Delivery of Information Across the Network
- Switching and Routing
- Translation Between Protocols
- Network Level Security

■ **Reliable transmission of data**



Layer 3: Edge Computing Layer

- Data reduction, filtering, and processing



Layer 3: Edge Computing Layer

- Edge computing is often referred to as the foglayer .
- At this layer, the emphasis is on data reduction and converting network data flows into information that is ready for storage and processing by higher layers.
- One of the basic principles of this reference model is that information processing is initiated as early and as close to the edge of the network as possible.
- Another important function that occurs at Layer 3 is the evaluation of data to see if it can be filtered or aggregated before being sent to a higher layer.
- This also allows for data to be reformatted or decoded, making additional processing by other systems easier.
- Thus, a critical function is assessing the data to see if predefined thresholds are crossed and any action or alerts need to be sent

Upper Layers: Layers 4–7

- The upper layers deal with handling and processing the IoT data generated by the bottom layer.

IoT Reference Model Layer	Functions
Layer 4: Data accumulation layer	Captures data and stores it so it is usable by applications when necessary. Converts event-based data to query-based processing.
Layer 5: Data abstraction layer	Reconciles multiple data formats and ensures consistent semantics from various sources. Confirms that the data set is complete and consolidates data into one place or multiple data stores using virtualization.
Layer 6: Applications layer	Interprets data using software applications. Applications may monitor, control, and provide reports based on the analysis of the data.
Layer 7: Collaboration and processes layer	Consumes and shares the application information. Collaborating on and communicating IoT information often requires multiple steps, and it is what makes IoT useful. This layer can change business processes and delivers the benefits of IoT.

Summary of Layers 4–7 of the IoTWF Reference Model

IT and OT Responsibilities in the IoT Reference Model

➤ In the IoT Reference Model, IT (Information Technology) and OT (Operational Technology) responsibilities intersect and overlap.

OT (Operational Technology) typically focuses on:

- 1. Managing and controlling physical devices, sensors, and actuators.
- 2. Ensuring the operation and maintenance of industrial equipment and processes.
- 3. Focusing on reliability, availability, and safety.

IT (Information Technology) typically focuses on:

- 1. Data management, processing, and analytics.
- 2. Application development and integration.
- 3. Ensuring data security, privacy, and governance.

In IoT, IT and OT converge, requiring collaboration and alignment between these two domains.

This convergence enables:- Better decision-making through data-driven insights- Improved operational efficiency and automation- Enhanced innovation and business opportunities

The IoT Reference Model helps bridge the gap between IT and OT, facilitating a more integrated and effective approach to IoT solution development and deployment.

IT and OT Responsibilities in the IoT Reference Model

❑ IT Responsibilities

- Data Management: Storage, processing, and analysis of data generated by devices.
- Applications: Development and deployment of applications that utilize IoT data.

❑ OT Responsibilities

- Device Management: Monitoring, control, and management of devices and sensors.
- Operational Processes: Management of operational processes and workflows.

Key Differences between IT and OT

- Focus: IT focuses on data, while OT focuses on devices and operational processes.
- Functionality: IT handles data processing and analysis, while OT handles device control and monitoring.

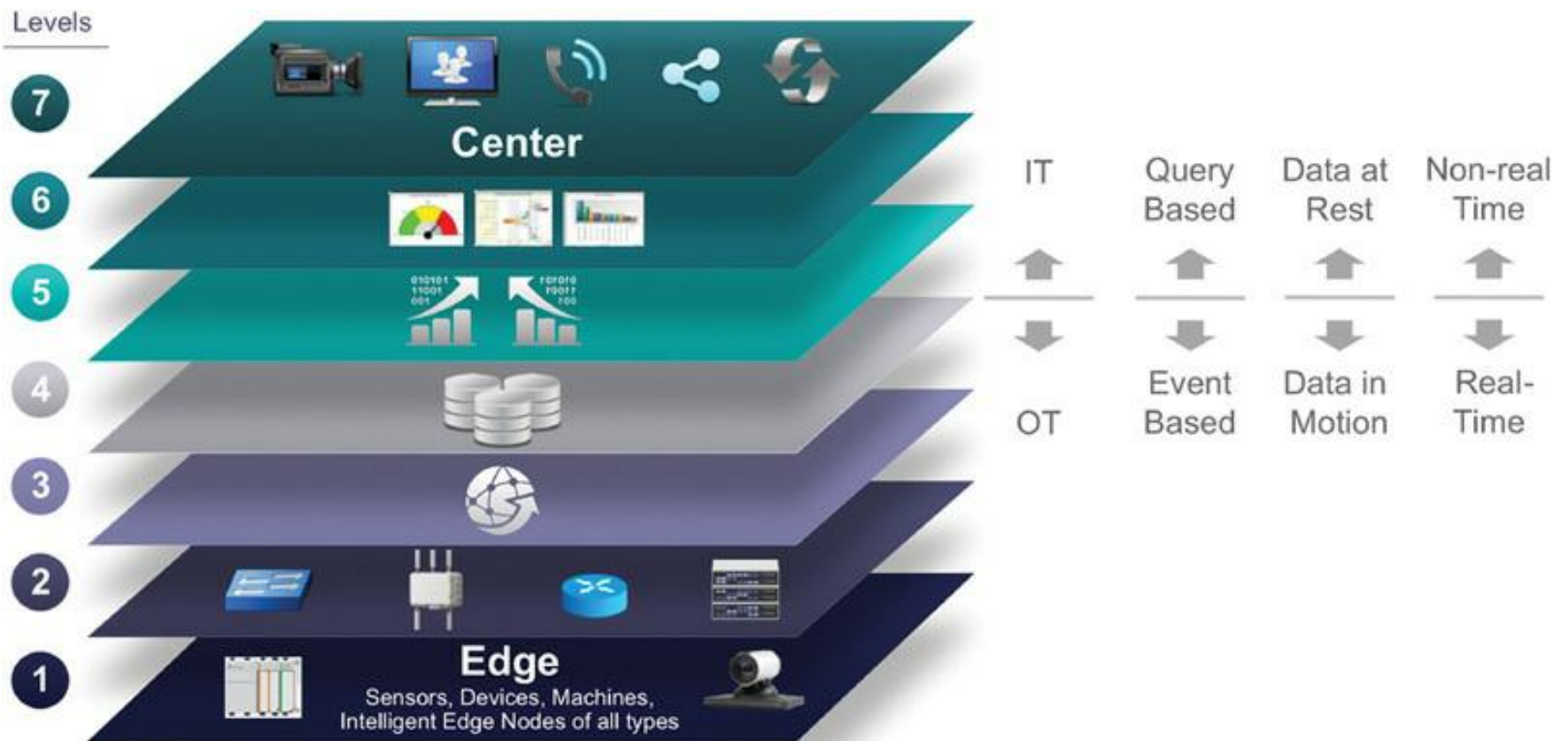


Fig: IoT Reference Model Separation of IT and OT

Benefits of IoTWF Reference Model

- Decompose IoT problems: Break down IoT issues into smaller parts.
- Identify technologies: Determine different technologies at each layer and their relationships.
- Multi-vendor support: Enable different vendors to provide different parts of the system.
- Interoperability: Define interfaces for seamless interaction between components.
- Tiered security: Implement a security model enforced at transition points between levels.

Alternative IoT Models

These are the models endorsed by various organizations and standards bodies and are often specific to certain industries or IoT applications.

IoT Reference Model	Description
Purdue Model for Control Hierarchy	The Purdue Model for Control Hierarchy (see www.cisco.com/c/en/us/td/docs/solutions/Verticals/EttF/EttFDIG/ch2_EttF.pdf) is a common and well-understood model that segments devices and equipment into hierarchical levels and functions. It is used as the basis for ISA-95 for control hierarchy, and in turn for the IEC-62443 (formerly ISA-99) cyber security standard. It has been used as a base for many IoT-related models and standards across industry. The Purdue Model's application to IoT is discussed in detail in Chapter 9, "Manufacturing," and in Chapter 10, "Oil & Gas."

Industrial Internet
Reference Architecture
(IIRA) by Industrial
Internet Consortium
(IIC)

The IIRA is a standards-based open architecture for Industrial Internet Systems (IISs). To maximize its value, the IIRA has broad industry applicability to drive interoperability, to map applicable technologies, and to guide technology and standard development. The description and representation of the architecture are generic and at a high level of abstraction to support the requisite broad industry applicability. The IIRA distills and abstracts common characteristics, features and patterns from use cases well understood at this time, predominantly those that have been defined in the IIC.

For more information, see www.iiconsortium.org/IIRA.htm.

Internet of Things–
Architecture (IoT-A)

IoT-A created an IoT architectural reference model and defined an initial set of key building blocks that are foundational in fostering the emerging Internet of Things. Using an experimental paradigm, IoT-A combined top-down reasoning about architectural principles and design guidelines with simulation and prototyping in exploring the technical consequences of architectural design choices.

For more information, see <https://vdivde-it.de/en>.

A Simplified IoT Architecture

The IoT architecture consists of two parallel stacks:

1. Core IoT Functional Stack: Things (sensors, devices), Communication Network & Applications
2. IoT Data Management and Compute Stack: Edge Computing, Fog Computing & Cloud Computing.

This simplified architecture separates core IoT functionality from data management, enabling industry-specific use cases and efficient data processing.

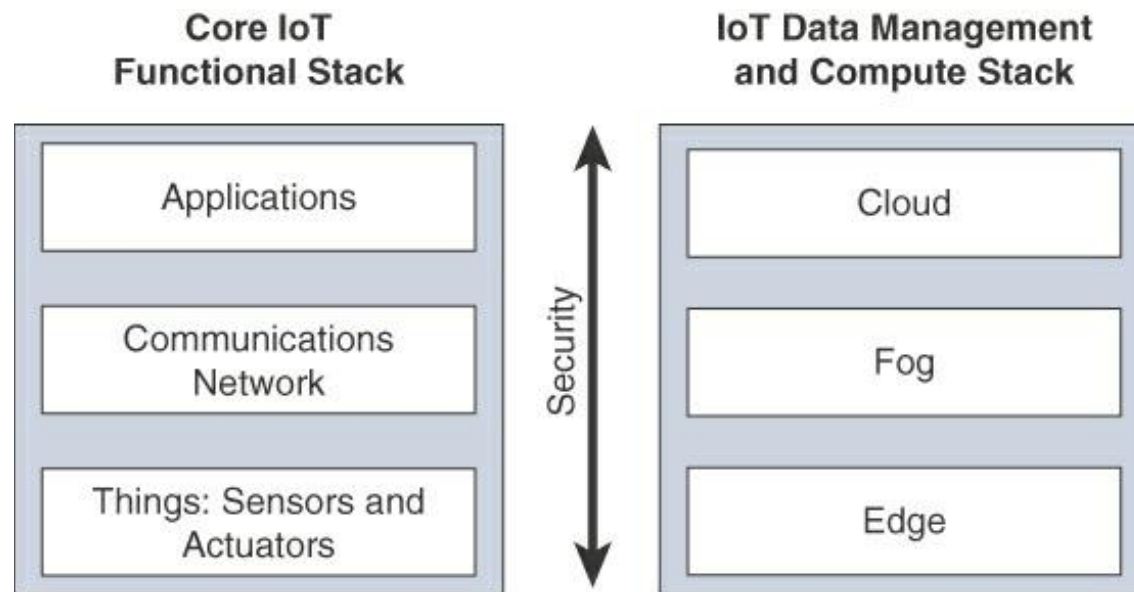


Fig: Simplified IoT Architecture

A Simplified IoT Architecture:

- The framework separates the core IoT and data management into parallel and aligned stacks, allowing us to carefully examine the functions of both the network and the applications at each stage of a complex IoT system.
- This separation gives us better visibility into the functions of each layer.
- The network communications layer of the IoT stack itself involves a significant amount of detail and incorporates a vast array of technologies.
- Consider for a moment the heterogeneity of IoT sensors and the many different ways that exist to connect them to a network.
- The network communications layer needs to consolidate these together, offer gateway and backhaul technologies, and ultimately bring the data back to a central location for analysis and processing.
- Many of the last-mile technologies used in IoT are chosen to meet the specific requirements of the endpoints and are unlikely to ever be seen in the IT domain.
- However, the network between the gateway and the data center is composed mostly of traditional technologies that experienced IT professionals would quickly recognize.

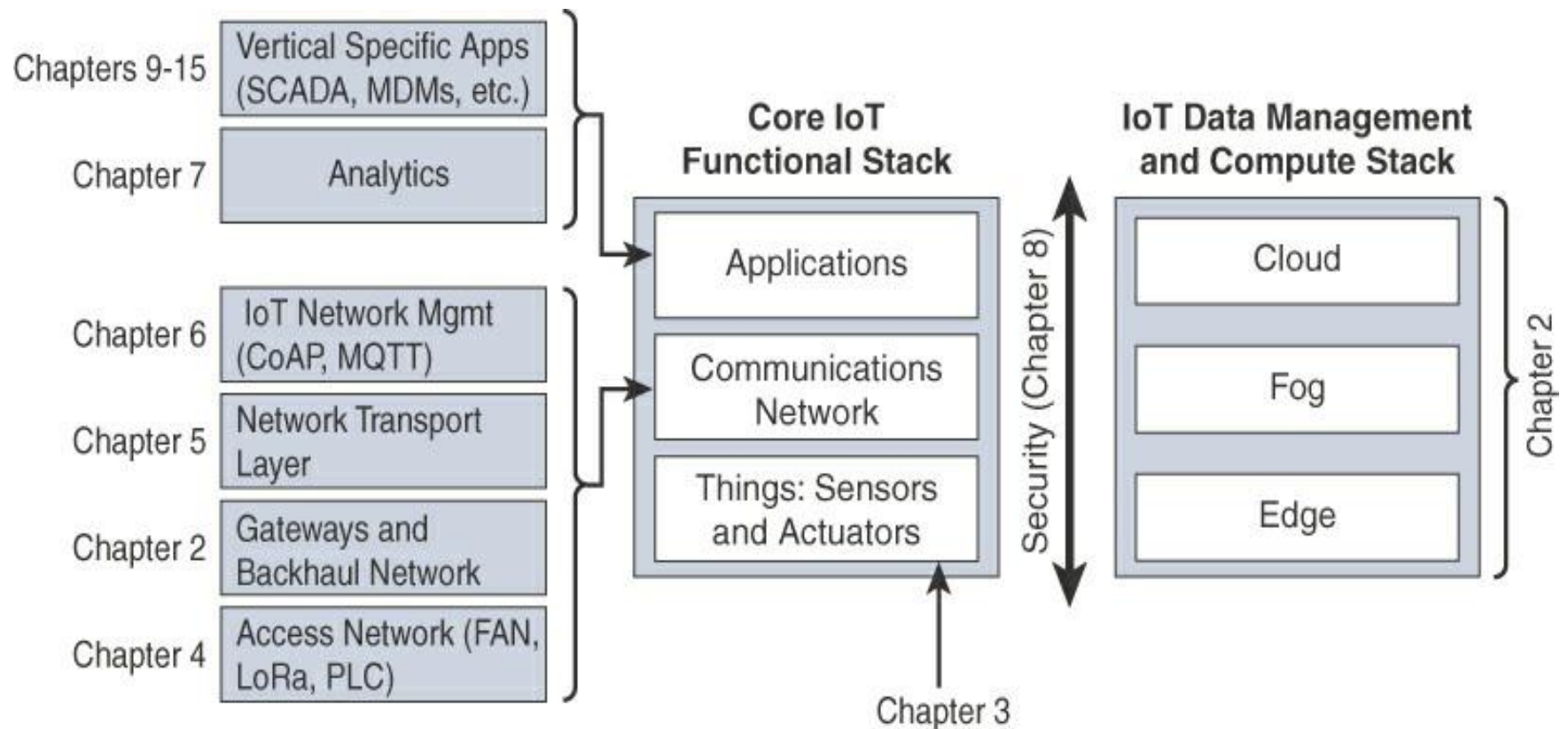


Fig: Expanded View of the Simplified IoT Architecture

A Simplified IoT Architecture:

- In the model presented, data management is aligned with each of the three layers of the Core IoT Functional Stack.
- The three data management layers are the edge layer (data management within the sensors themselves), the fog layer (data management in the gateways and transit network), and the cloud layer (data management in the cloud or central data center).
- The Core IoT Functional Stack can be expanded into sublayers containing greater detail and specific network functions.
- For example, the communications layer is broken down into four separate sublayers: the access network, gateways and backhaul, IP transport, and operations and management sublayers.
- The applications layer of IoT networks is quite different from the application layer of a typical enterprise network.
- IoT often involves a strong big data analytics component.
- IoT is not just about the control of IoT devices but, rather, the useful insights gained from the data generated by those devices.
- Thus, the applications layer typically has both analytics and industry-specific IoT control system components.

Core IoT Functional Stack

- **Layer 1:** Things: Sensors and Actuators Layer: Sensors and devices.
- **Layer 2:** Communications Network Layer: Connecting devices to the
- **Layer 3:** Applications and Analytics Layer: Analytics and industry-specific control systems.

- IoT networks are built around the concept of things, or smart objects performing functions and delivering new connected services.
- These objects are smart because they use a combination of contextual information and configured goals to perform actions.
- These actions can be self-contained (that is, the smart object does not rely on external systems for its actions); however, in most cases, the thing interacts with an external system to report information that the smart object collects, to exchange with other objects, or to interact with a management platform.
- In this case, the management platform can be used to process data collected from the smart object and also guide the behavior of the smart object.
- From an architectural standpoint, several components have to work together for an IoT network to be operational:

Layer 1: Things: Sensors and Actuators Layer, **Layer 2:** Communications Network Layer
Layer 3: Applications and Analytics Layer

1. Things layer: At this layer, the physical devices need to fit the constraints of the environment in which they are deployed while still being able to provide the information needed.

- Smart objects (sensors, actuators) with different characteristics:
- Battery-powered or power-connected
- Mobile or static
- Low or high reporting frequency
- Simple or rich data
- Report range
- Object density per cell
- Mobility and throughput requirements

2. Communication Network Layer: Enables communication between smart objects and external environments:

- Access Network Sub-layer: Wireless technologies (802.11ah, 802.15.4g, LoRa) and wired networks
- Gateways and Backhaul Sub-layer: Forwards data to central servers, acts as routers
- Network Transport Sub-layer: Uses transport layer protocols (IP, UDP)
- IoT Network Management Sub-layer: Accesses and exchanges data with sensors and applications (CoAP, MQTT)

Communications network layer: When smart objects are not self contained, they need to communicate with an external system.

➤ This communication uses a wireless technology. This layer has four sublayers:

Access network sublayer:

- The last mile of the IoT network is the access network.
- This is made up of wireless technologies such as 802.11ah, 802.15.4g, and LoRa.
- The sensors connected to the access network may also be wired.

Gateways and backhaul network sublayer:

- A common communication system organizes multiple smart objects in a given area around a common gateway.
- The gateway communicates directly with the smart objects.
- The role of the gateway is to forward the collected information through a longer-range medium (called the backhaul) to a headend central station where the information is processed.
- This information exchange is an application function, which is the reason this object is called a gateway.
- On IP networks, this gateway also forwards packets from one IP network to another, and it therefore acts as a router.

Network transport sublayer:

- For communication to be successful, network and transport layer protocols such as IP and UDP must be implemented to support the variety of devices to connect and media to use.

IoT network management sublayer:

- Additional protocols must be in place to allow the headend applications to exchange data with the sensors.
- Examples include CoAP and MQTT.

3. Application and analytics layer:

- At the upper layer, an application needs to process the collected data, not only to control the smart objects when necessary, but to make intelligent decision based on the information collected and, in turn, instruct the things or other systems to adapt to the analyzed conditions and change their behaviors or parameters.
- Analyzes data, makes decisions, and instructs objects to change behavior.
- Access Network Technologies
 - PAN (Bluetooth)
 - HAN (Zigbee)
 - NAN (802.11ah)
 - FAN (Wireless HART)
 - LAN (IEEE 802.11)
 - MAN (Wi-Fi)
 - WAN (Internet)

IoT Network Topologies and Layers

- Point-to-Point Topology: Communication between two nodes
- Point-to-MultiPoint Topology: One device communicates with multiple devices
- Topologies: Star, Mesh, and others, with Coordinator nodes (FFD) and Reduced Function Devices (RFD)
- Communication Network Layer
- Gateways and Backhaul: Transfers data from sensors to center, using technologies like Wi-Fi and WiMax- Network Transport Layer: Uses IP, UDP, and TCP protocols for secure communication
- IoT Network Management Layer: Uses HTTP, MQTT, XMPP, and CoAP protocols for data transfer and management

Applications and Analytics

- Analytics vs Control Applications: Analytics applications collect and analyze data, while control applications control smart devices
- Data vs Network Analytics: Data analytics monitors sensor data, while network analytics monitors network performance
- Smart Services: Intelligent services provided by objects, such as smart home automation and industrial monitoring

IoT Data Management and Compute Stack

The massive scale of IoT networks drives new architectures to manage the vast amount of data generated by IoT devices.

IoT Data Management- Data Collection: Sensors generate data, collected by meter management systems, and analyzed in the cloud

Challenges

- Data Volume: IoT devices generate enormous amounts of data.
- Data Variety: Much of the data is unstructured and of little use on its own.
- Latency: Milliseconds matter in many IoT applications.
- Bandwidth: Limited bandwidth in last-mile IoT networks.

Cloud Computing in IoT- Centralized Architecture: Cloud computing model for IoT data analysis- Data-Related Issues:

- Limited bandwidth
- High latency
- Expensive backhaul links
- High volume of data
- Real-time data management challenges

Fog Computing

- Fog Computing is an architecture that uses edge devices to perform computing, storage, and communication and acts over the internet This helps IoT to work efficiently.
- This involves four layers the bottom most layer has the end points(smart objects-sensors(low power and low bandwidth)).the next layer is the edge layer or fog layer, core IPV6 Network layer for communication to the data center or cloud.
- Characteristics of fog computing
 - Fog nodes provide contextual awareness.
 - Fog nodes provide services and application on wide geographic distribution.
 - In gateway router 4000 fog nodes can be deployed near IoT endpoints.
 - Fog computing provides wireless access to all end points.
 - This computing provides real time interactions and pre processing of data.

Benefits:

- Minimizes latency
- Offloads network traffic
- Keeps sensitive data local

Fog Nodes: Devices with computing, storage, and network connectivity.

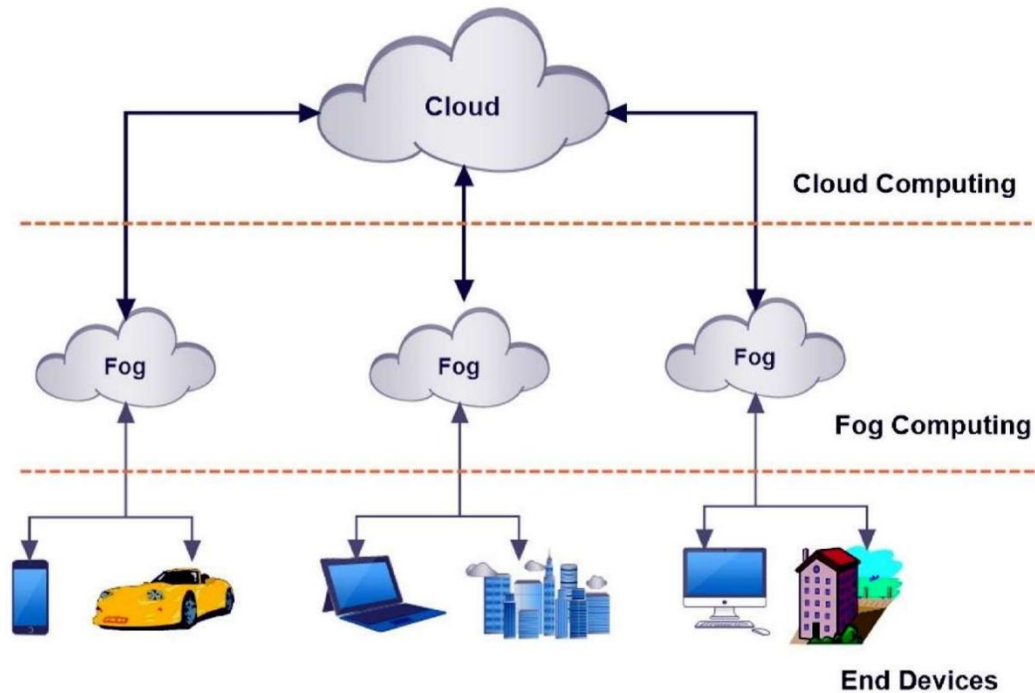


Fig: Fog Computing

Fog Computing

Distributed data management and computing at the edge of the network.

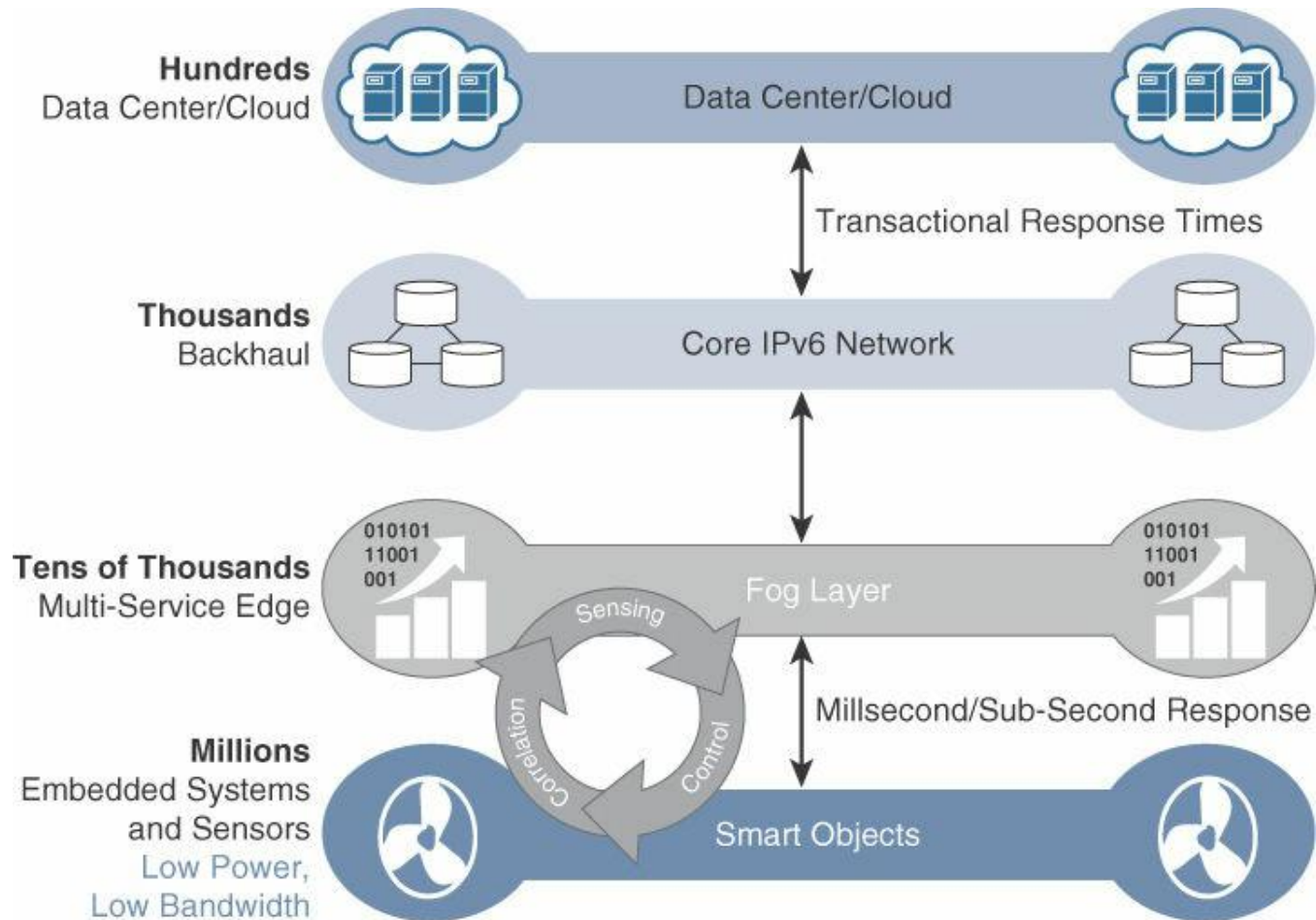


Fig: The IoT Data Management and Compute Stack with Fog Computing

Edge Computing (Mist Computing)

- Edge computing enables connected devices to process the data where it was generated called the “edge.”
- This can be done on the device itself (i.e. sensors), or very near to the device, and provides an alternative way of sending data to a central cloud for further processing.
- Features of Edge computing:
 - Edge computing is called mist computing.
 - Edge computing extends to the end device itself.
 - Effective data processing.
 - Reduces the consumption internet bandwidth.
 - It provides security to the sensitive data

Computing and data management within IoT devices or sensors.

Benefits:

Real-time analysis and response

- Reduced data transmission
- Improved efficiency

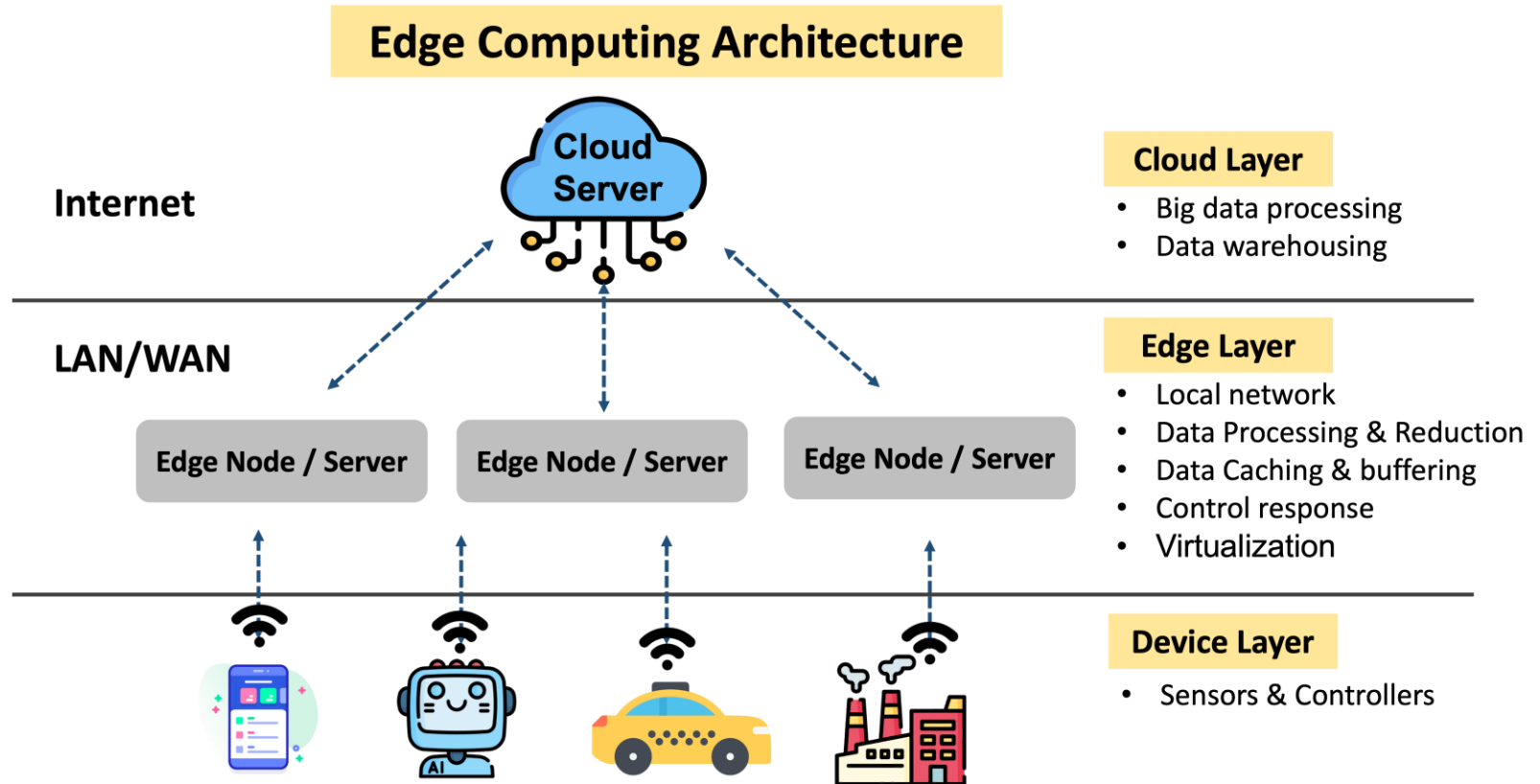


Figure : Edge computing architecture overview
Source : The research team

Fig: Edge Computing

The Hierarchy of Edge, Fog, and Cloud

- Edge or fog computing in no way replaces the cloud but they complement each other, and many use cases actually require strong cooperation between layers.
- Edge and fog computing layers simply act as a first line of defense for filtering, analyzing, and otherwise managing data endpoints.
- This saves the cloud from being queried by each and every node for each event.
- This model suggests a hierarchical organization of network, compute, and data storage resources.
- At each stage, data is collected, analyzed, and responded to when necessary, according to the capabilities of the resources at each layer.
- As data needs to be sent to the cloud, the latency becomes higher.
- The advantage of this hierarchy is that a response to events from resources close to the end device is fast and can result in immediate benefits, while still having deeper compute resources available in the cloud when necessary.

The Hierarchy of Edge, Fog, and Cloud

- heterogeneity of IoT devices also means a heterogeneity of edge and fog computing resources.
- While cloud resources are expected to be homogenous, it is fair to expect that in many cases both edge and fog resources will use different operating systems, have different CPU and data storage capabilities, and have different energy consumption profiles.
- Edge and fog thus require an abstraction layer that allows applications to communicate with one another.
- The abstraction layer exposes a common set of APIs for monitoring, provisioning, and controlling the physical resources in a standardized way.
- The abstraction layer also requires a mechanism to support virtualization, with the ability to run multiple operating systems or service containers on physical devices to support multitenancy and application consistency across the IoT system.

The Hierarchy of Edge, Fog, and Cloud

- Edge: Real-time analysis and response.
- Fog: Distributed computing and data management.
- Cloud: Historical analysis, big data analytics, and long-term storage.

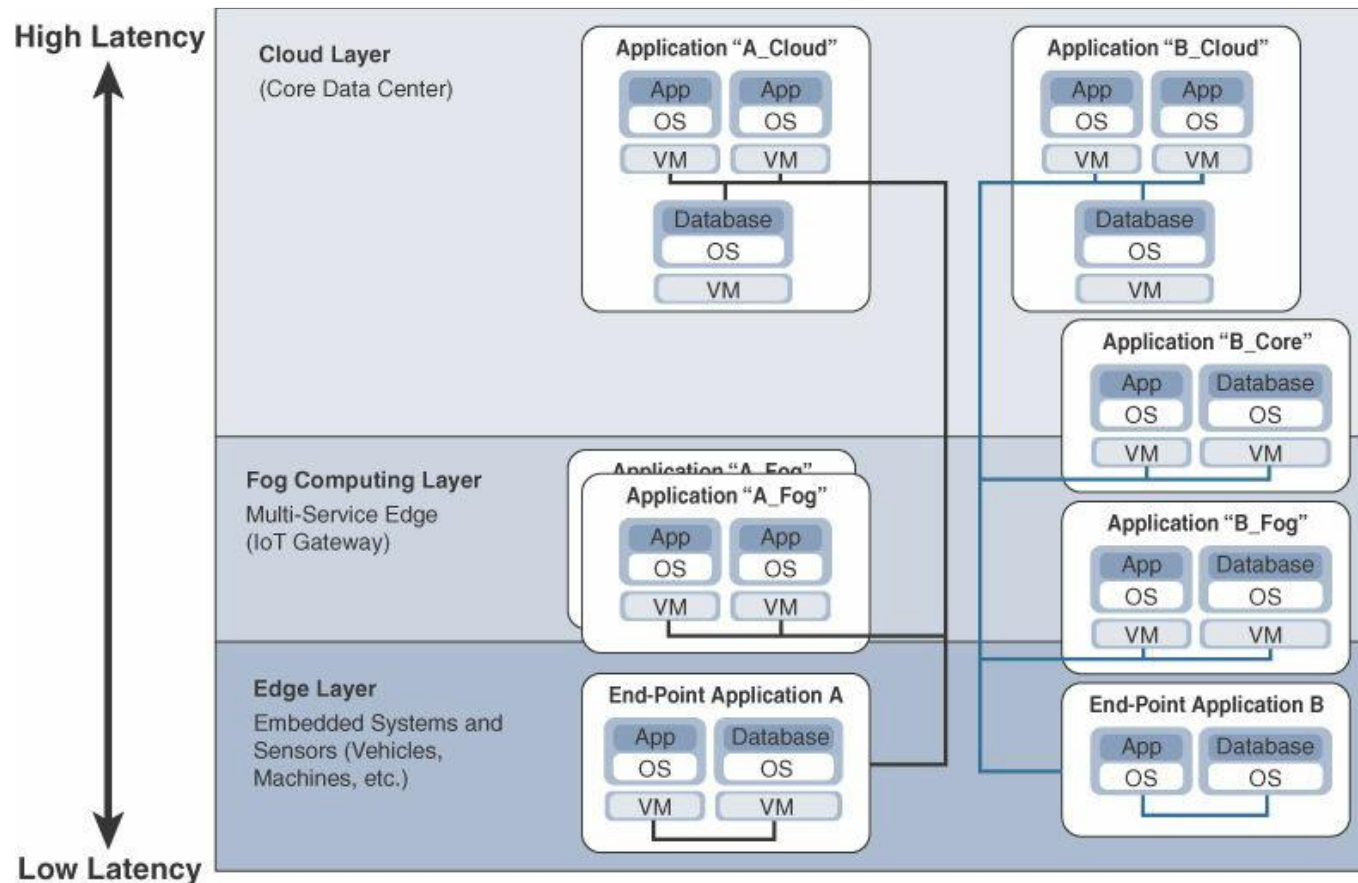


Fig: Distributed Compute and Data Management Across an IoT System

Sensors in IoT

Sensors are devices that measure physical quantities and convert them into digital representations. They play a crucial role in IoT systems, enabling devices to perceive and respond to their environment.

Types of Sensors

1. Active or Passive: Active sensors produce energy output, while passive sensors receive energy.
2. Invasive or Non-invasive: Invasive sensors are part of the environment, while non-invasive sensors are external.
3. Contact or No-contact: Contact sensors require physical contact, while no-contact sensors do not.
4. Absolute or Relative: Absolute sensors measure on an absolute scale, while relative sensors measure differences.

Sensor Types	Description	Examples
Position	A position sensor measures the position of an object; the position measurement can be either in absolute terms (absolute position sensor) or in relative terms (displacement sensor). Position sensors can be linear, angular, or multi-axis.	Potentiometer, inclinometer, proximity sensor
Occupancy and motion	Occupancy sensors detect the presence of people and animals in a surveillance area, while motion sensors detect movement of people and objects. The difference between the two is that occupancy sensors generate a signal even when a person is stationary, whereas motion sensors do not.	Electric eye, radar
Velocity and acceleration	Velocity (speed of motion) sensors may be linear or angular, indicating how fast an object moves along a straight line or how fast it rotates. Acceleration sensors measure changes in velocity.	Accelerometer, gyroscope
Force	Force sensors detect whether a physical force is applied and whether the magnitude of force is beyond a threshold.	Force gauge, viscometer, tactile sensor (touch sensor)
Pressure	Pressure sensors are related to force sensors, measuring force applied by liquids or gases. Pressure is measured in terms of force per unit area.	Barometer, Bourdon gauge, piezometer
Flow	Flow sensors detect the rate of fluid flow. They measure the volume (mass flow) or rate (flow velocity) of fluid that has passed through a system in a given period of time.	Anemometer, mass flow sensor, water meter

Acoustic	Acoustic sensors measure sound levels and convert that information into digital or analog data signals.	Microphone, geophone, hydrophone
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass. Humidity levels can be measured in various ways: absolute humidity, relative humidity, mass ratio, and so on.	Hygrometer, humistor, soil moisture sensor
Light	Light sensors detect the presence of light (visible or invisible).	Infrared sensor, photodetector, flame detector
Radiation	Radiation sensors detect radiation in the environment. Radiation can be sensed by scintillating or ionization detection.	Geiger-Müller counter, scintillator, neutron detector
Temperature	Temperature sensors measure the amount of heat or cold that is present in a system. They can be broadly of two types: contact and non-contact. Contact temperature sensors need to be in physical contact with the object being sensed. Non-contact sensors do not need physical contact, as they measure temperature through convection and radiation.	Thermometer, calorimeter, temperature gauge
Chemical	Chemical sensors measure the concentration of chemicals in a system. When subjected to a mix of chemicals, chemical sensors are typically selective for a target type of chemical (for example, a CO ₂ sensor senses only carbon dioxide).	Breathalyzer, olfactometer, smoke detector
Biosensors	Biosensors detect various biological elements, such as organisms, tissues, cells, enzymes, antibodies, and nucleic acid.	Blood glucose biosensor, pulse oximetry, electrocardiograph

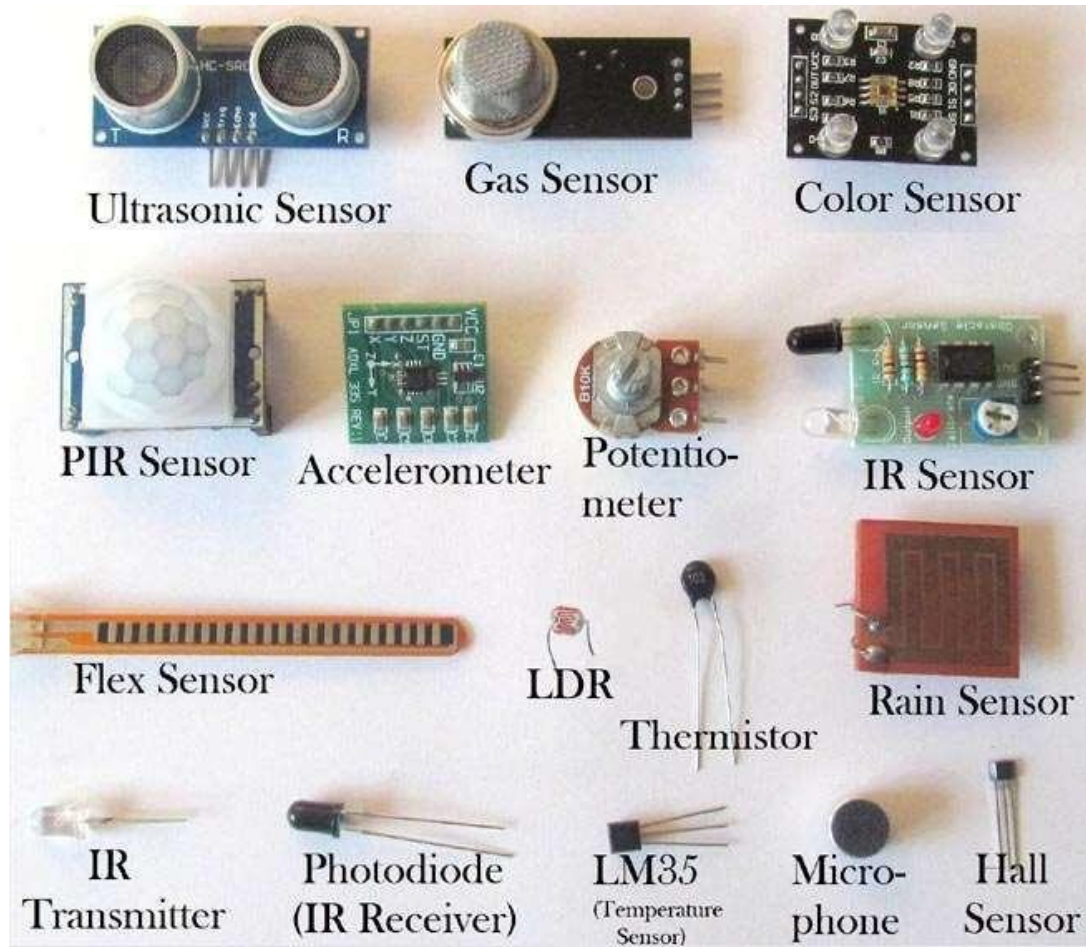
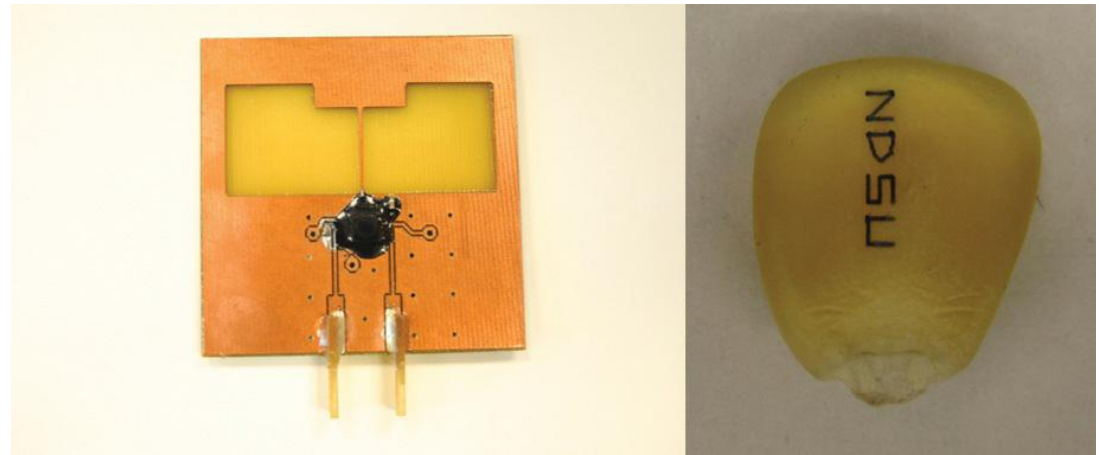


Fig: various types of sensors

Sensor Applications

- Precision Agriculture: Sensors measure soil characteristics, such as pH levels, moisture, and nutrient levels.
- Smart Homes: Sensors detect temperature, humidity, motion, and more.
- Intelligent Vehicles: Sensors monitor speed, pressure, temperature, and other factors.
- Connected Cities: Sensors track traffic, weather, and environmental conditions.



Actuators in IoT

Actuators are devices that receive control signals and trigger physical effects, such as motion or force. They complement sensors, enabling IoT systems to interact with and impact the physical world.

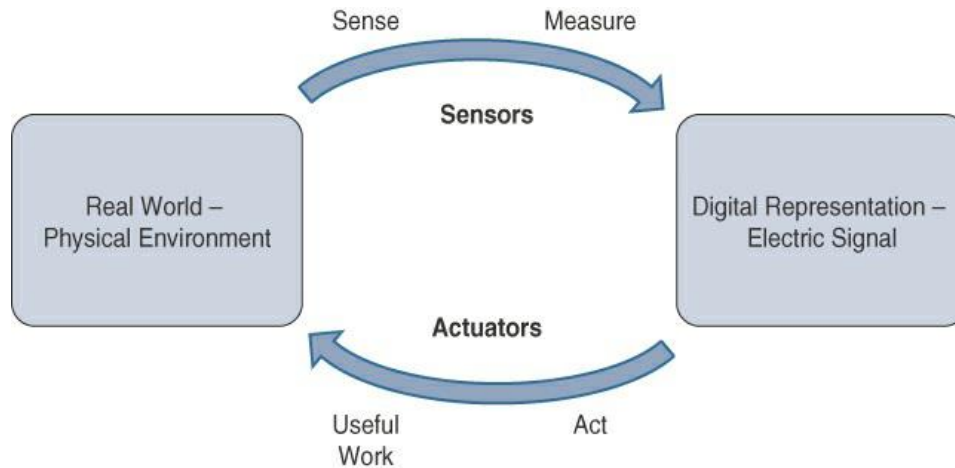


Fig: How Sensors and Actuators Interact with the Physical World

Types of Actuators

1. Classification by Energy Type: Actuators can be categorized based on their energy source, such as electric, pneumatic, hydraulic, or thermal.
2. Classification by Motion: Actuators can be classified based on the type of motion they produce, such as linear, rotary, or multi-axis.

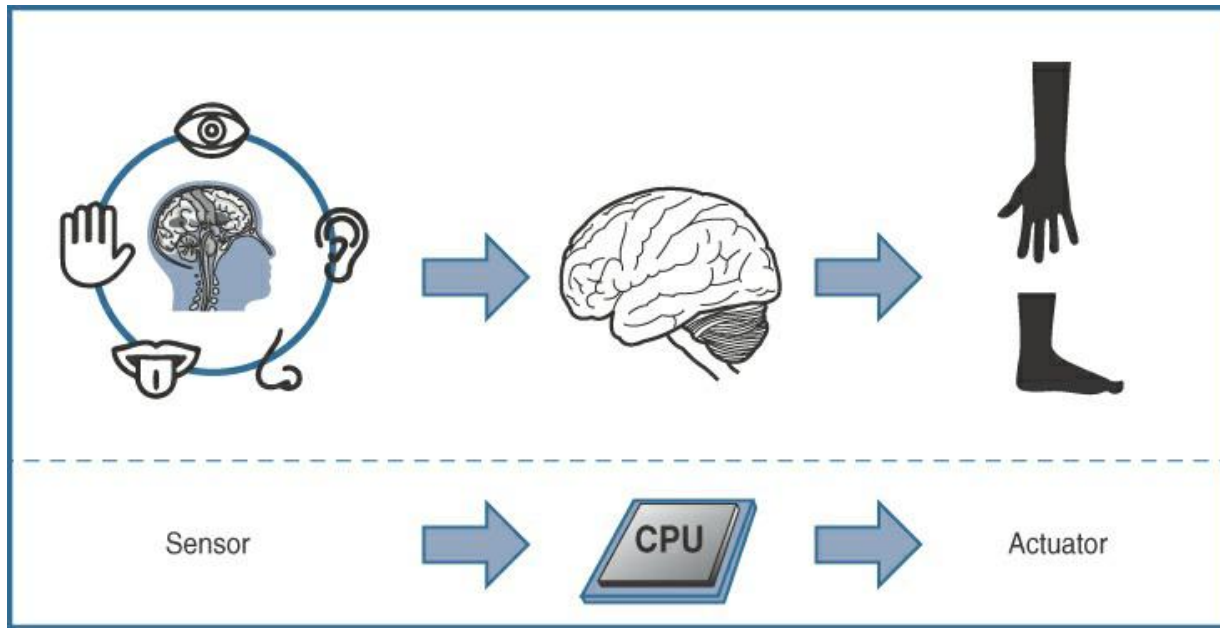


Fig: Comparison of Sensor and Actuator Functionality with Humans

Actuator Applications

1. Precision Agriculture: Actuators can control valves to deliver optimized amounts of water, pesticides, fertilizers, and herbicides based on sensor readings.
2. Industrial Automation: Actuators can perform tasks such as assembly, inspection, and material handling.
3. Robotics: Actuators enable robots to move and interact with their environment.

Type	Examples
Mechanical actuators	Lever, screw jack, hand crank
Electrical actuators	Thyristor, biopolar transistor, diode
Electromechanical actuators	AC motor, DC motor, step motor
Electromagnetic actuators	Electromagnet, linear solenoid
Hydraulic and pneumatic actuators	Hydraulic cylinder, pneumatic cylinder, piston, pressure control valves, air motors
Smart material actuators (includes thermal and magnetic actuators)	Shape memory alloy (SMA), ion exchange fluid, magnetorestrictive material, bimetallic strip, piezoelectric bimorph
Micro- and nanoactuators	Electrostatic motor, microvalve, comb drive

Fig: Actuator Classification by Energy Type

Smart Objects

Smart objects are the building blocks of IoT, transforming everyday objects into intelligent, networked devices that can learn from and interact with their environment.

Characteristics of Smart Objects

1. **Processing Unit:** A smart object has a processing unit for acquiring, processing, and analyzing data.
2. **Sensor(s) and/or Actuator(s):** Smart objects interact with the physical world through sensors and actuators.
3. **Communication Device:** Smart objects have a communication unit for connecting with other devices and the outside world.
4. **Power Source:** Smart objects require a power source, often with limited power consumption.

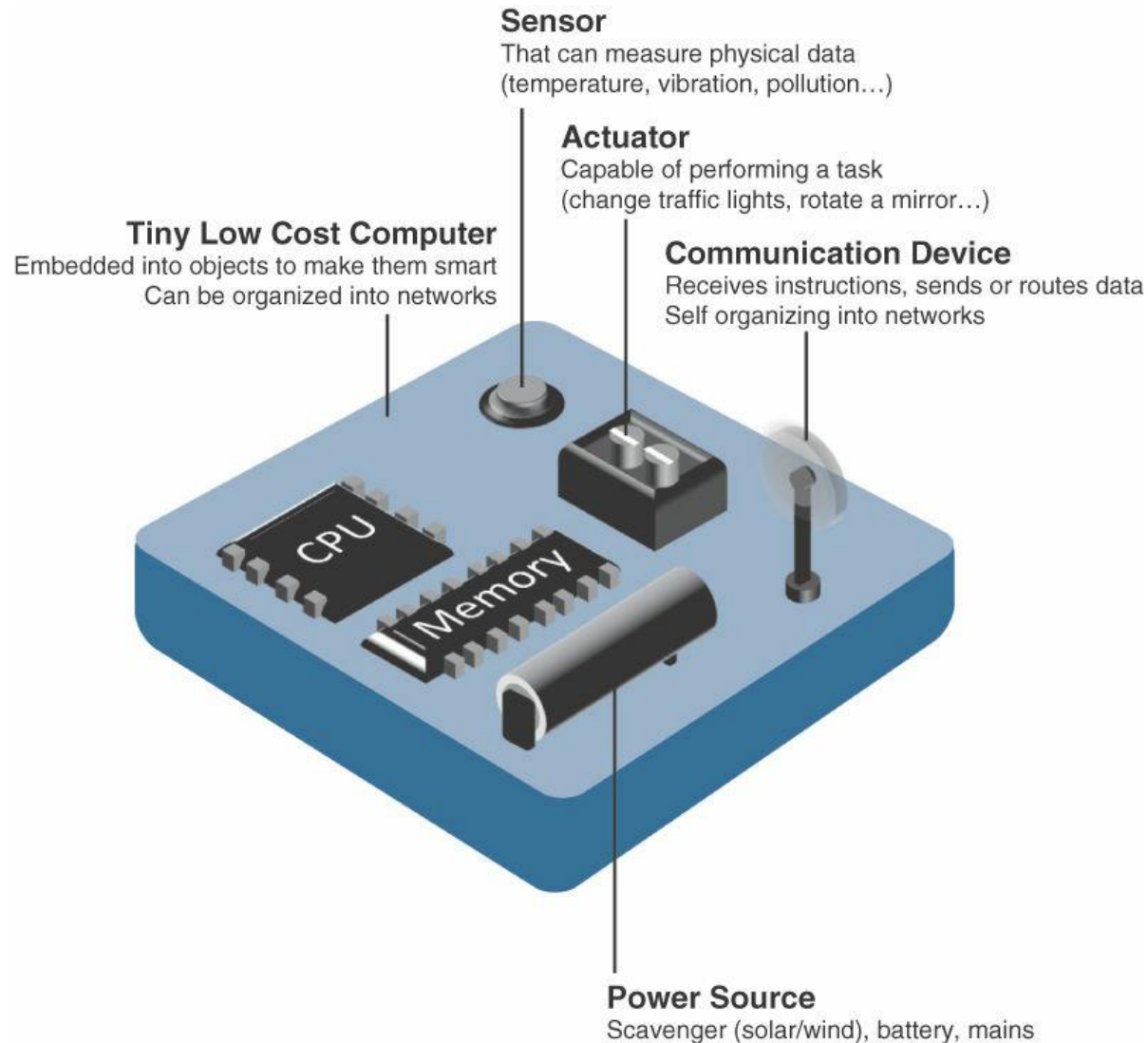


Fig: Characteristics of a Smart Object

Trends in Smart Objects

1. Decreasing Size: Smart objects are getting smaller, making them easier to embed in everyday objects.
2. Decreasing Power Consumption: Smart objects are becoming more power-efficient, with some lasting 10+ years on a single battery.
3. Increasing Processing Power: Processors are getting more powerful and smaller, enabling more complex tasks.
4. Improving Communication Capabilities: Wireless speeds and ranges are increasing, enabling more sophisticated applications.

Wireless Sensor Networks (WSNs)

WSNs are networks of wirelessly connected smart objects that can sense and measure their environment.

They offer advantages such as:

1. Greater Deployment Flexibility: WSNs can be deployed in harsh environments and hard-to-reach places.
2. Simpler Scaling: WSNs can scale to large numbers of nodes with ease.
3. Lower Implementation Costs: WSNs can reduce implementation costs compared to wired networks.

Challenges in WSNs

1. Limited Processing Power: Smart objects in WSNs often have limited processing power and memory.
2. Lossy Communication: WSNs can experience packet loss and interference.
3. Limited Power: Smart objects in WSNs often have limited power sources, requiring efficient power management.

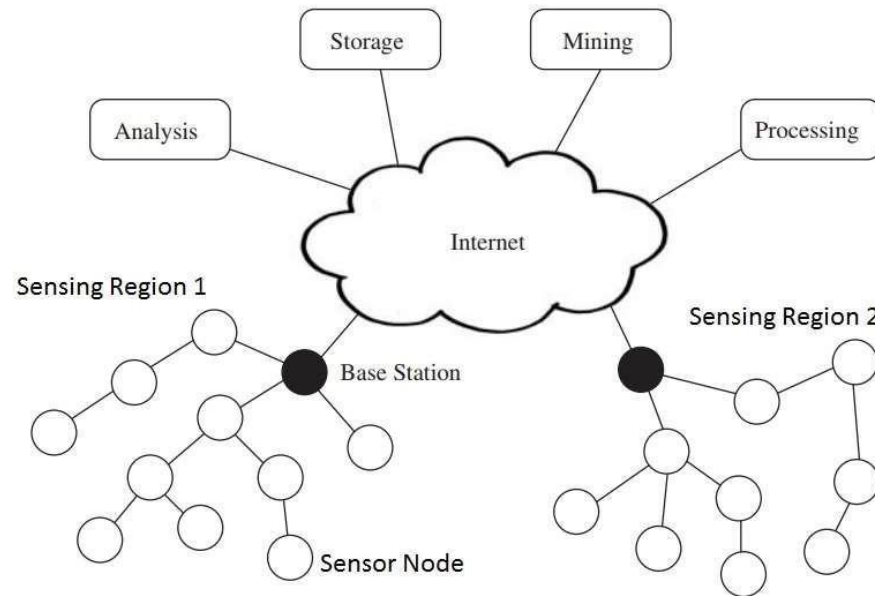


Fig: Wireless Sensor Networks (WSNs)

Applications of WSNs

- Smart Homes: WSNs can control temperature, lighting, and security systems.
- Industrial Automation: WSNs can monitor and control industrial processes.
- Environmental Monitoring: WSNs can detect earthquakes, forest fires, and other environmental phenomena.

Connecting Smart Objects

When connecting smart objects, several key criteria must be considered:

1. Range: Signal propagation and distance are crucial factors.
2. Frequency Bands: Licensed and unlicensed spectrum, including sub-GHz frequencies, are used for IoT connectivity.
3. Power Consumption: Devices may be powered or battery-powered, affecting connectivity options.
4. Topology: Star, mesh, and peer-to-peer topologies are common in IoT networks.
5. Constrained Devices: Devices with limited resources impact networking capabilities.
6. Constrained-Node Networks: Networks connecting smart objects may be low-power and lossy.

IoT Access Technologies

Several technologies are used to connect smart objects:

- 1.IEEE 802.15.4: A wireless protocol for low-power, low-data-rate applications.
- 2.IEEE 802.15.4g: An extension of 802.15.4 for smart utility networks.
- 3.IEEE 802.15.4e: An amendment to 802.15.4 for industrial and commercial applications.
- 4.IEEE 1901.2a: A technology for narrowband power line communications.
- 5.IEEE 802.11ah: A Wi-Fi-based technology for low-power, low-bandwidth applications.
- 6.LoRaWAN: A wireless technology for long-range, low-power communications.
- 7.NB-IoT and LTE Variations: Cellular technologies for IoT applications.

Range

The distance over which a signal needs to be propagated, determining the area of coverage for a selected wireless technology.

Range Categories:

- Short Range: Up to tens of meters (e.g., IEEE 802.15.1 Bluetooth, IEEE 802.15.7 Visible Light Communications)
- Medium Range: Tens to hundreds of meters (e.g., IEEE 802.11 Wi-Fi, IEEE 802.15.4, IEEE 802.3 Ethernet)
- Long Range: Greater than 1 mile (e.g., cellular, LPWA, outdoor IEEE 802.11 Wi-Fi)

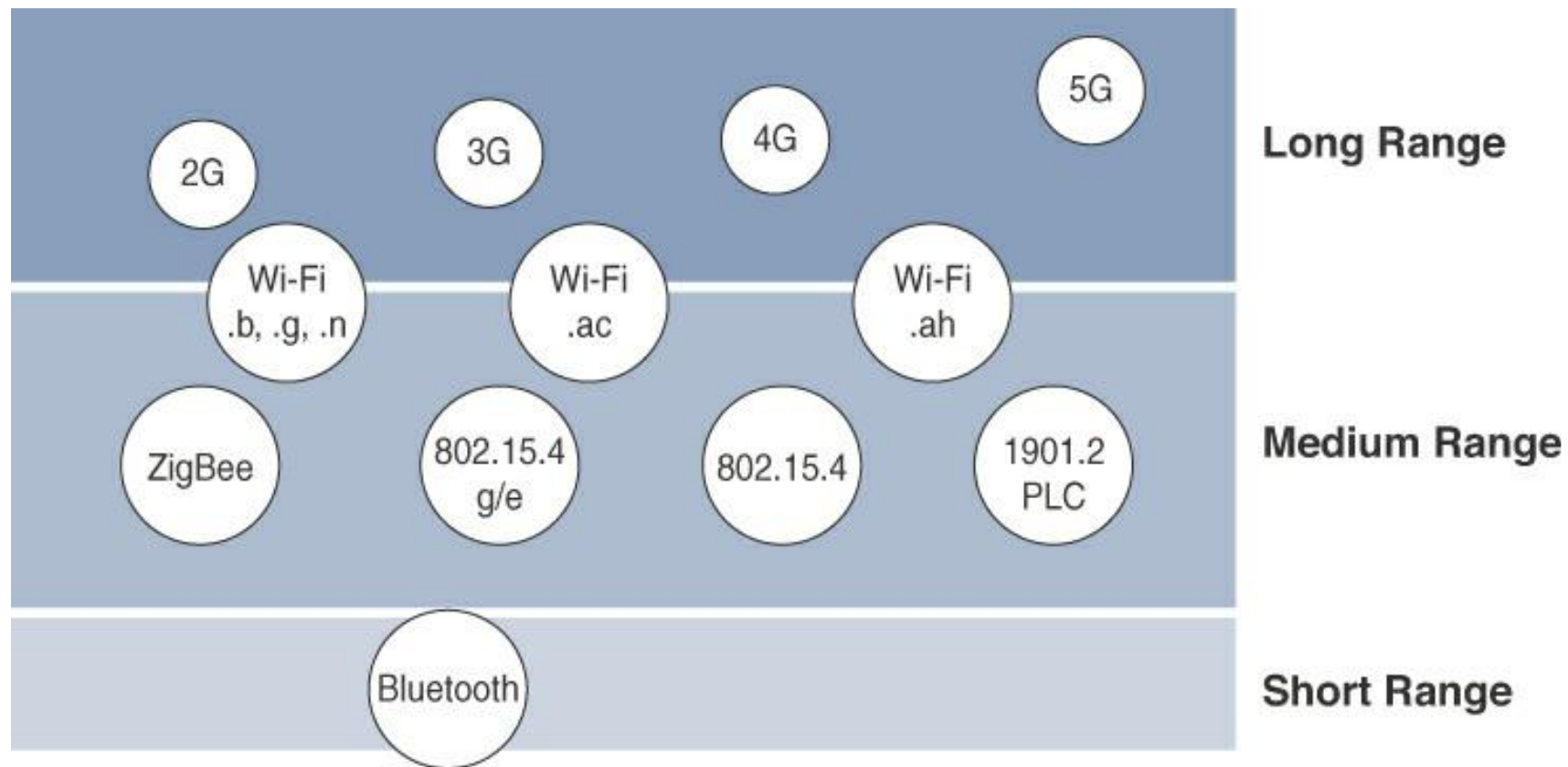


Fig: Wireless Access Landscape

THANK YOU